PHARMASENSE - AI-BASED PRESCRIPTION ANALYZER AND TREATMENT OUTCOME FORECASTING SYSTEM USING OCR, NLP, AND RANDOM FOREST MACHINE LEARNING ALGORITHM

A PROJECT REPORT

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The work embodied in the present project report entitled "**PHARMASENSE - AI-BASED PRESCRIPTION ANALYZER AND TREATMENT OUTCOME FORECASTING SYSTEM USING OCR, NLP, AND RANDOM FOREST MACHINE LEARNING ALGORITHM**" has been carried out by the students **SANTHOSH B, SRIHARI PRASAD S, SUBBIAH KARTHICK S.** The work reported herein is original and we declare that the project is their own work, except where specifically acknowledged, and has not been copied from other sources or been previously submitted for assessment.

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ABSTRACT

Pharma Sense is an intelligent healthcare assistance system designed to analyze medical prescriptions and forecast treatment outcomes using Artificial Intelligence. Using OCR technology, the system extracts textual information from the prescription, including medicine names, dosage instructions, and treatment duration. NLP techniques are then applied to identify key medical entities such as prescribed medicines, disease type, and recommended treatment period. if the patient fails to follow the prescribed medication plan. To enhance interpretability, the system also provides visual analytics using charts and graphs that compare doctor-prescribed treatment duration, current patient progress, and AI-optimized recovery predictions. These visualizations help patients clearly understand treatment efficiency and potential health outcomes. By combining OCR, NLP, and AI-powered reasoning, PharmaSense improves prescription understanding, promotes medication adherence, and provides intelligent health insights that assist patients in managing their treatment more effectively.

Keywords: Prescription Analysis, OCR (Optical Character Recognition), Natural Language Processing (NLP), Treatment Outcome Forecasting, Random Forest Machine Learning, Digital Health Records.

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SIGNATURE

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LIST OF SYMBOLS AND ABBREVIATIONS

OCR	-	OPTICAL CHARACTER RECOGNITION
RAM	-	RANDOM ACCESS MEMORY
SSD	-	SOLID STATE DRIVE
NLP	-	NATURAL LANGUAGE PROCESSING
VIT	-	VISION TRANSFORMER
NMC	-	NATIONAL MEDICAL COMMISSION
CSS	-	CASCADING STYLE SHEETS
NLTK	-	NATURAL LANGUAGE TOOLKIT
SPA CY	-	INDUSTRIAL-STRENGTH NATURAL LANGUAGE PROCESSING
MYSQL	-	MY STRUCTURED QUERY LANGUAGE
SQL	-	STRUCTURED QUERY LANGUAGE
JPG	-	JOINT PHOTOGRAPHIC EXPERTS GROUP
PNG	-	GRAPHICS PROCESSING UNIT

CHAPTER 1

INTRODUCTION

1.1 DESCRIPTION

PharmaSense is a comprehensive, AI-powered, full-stack telemedicine platform designed to bridge the significant gap between patients and National Medical Commission (NMC)-verified Indian doctors through a secure, intelligent, and highly accessible digital healthcare experience. In an era where digital transformation is reshaping every industry, healthcare remains one of the sectors most in need of technology-driven solutions that can simultaneously improve accessibility, reduce delays, enhance accuracy, and ensure the authenticity of medical services. PharmaSense was conceived and developed as a direct response to these needs, bringing together multiple advanced technologies under a single unified platform to serve patients, doctors, and healthcare administrators across India.

The PharmaSense system is built around two tightly integrated major subsystems that together form a complete end-to-end digital health solution. The first subsystem is the AI-based Prescription Analyser, which addresses the widespread challenge patients face when trying to understand complex medical prescriptions. The second subsystem is the Doctor–Patient Consultation Platform, which enables patients to find, connect with, and consult verified medical professionals through a structured and secure digital workflow. Both subsystems work in harmony, sharing the same user accounts, medical records, and data infrastructure, creating a seamless healthcare experience that would otherwise require multiple separate applications.

The AI Prescription Analyser addresses one of the most common problems in everyday healthcare the difficulty patients face when reading and understanding their own prescriptions. Medical prescriptions, whether handwritten by a doctor or printed from a hospital system, are typically filled with complex medical terminology, abbreviated drug names, Latin instructions, and dosage formats that are unfamiliar to

the average patient. The AI Prescription Analyser allows patients to upload their medical prescriptions in image format (JPG or PNG) or as PDF documents through a simple and intuitive interface. Once uploaded, the system forwards the document to the AI processing engine, which uses OpenAI's state-of-the-art multimodal language model to perform intelligent text extraction and natural language understanding on the prescription content.

1.2 DOMAIN SPECIFICATION

The PharmaSense system operates within the domain of Healthcare Technology and Medical Informatics, a rapidly evolving field that focuses on the application of digital technologies, computing systems, artificial intelligence, and data analytics to improve the quality, efficiency, and accessibility of healthcare services and patient outcomes. Healthcare informatics as a discipline combines elements of computer science, information science, and clinical medicine to create systems and tools that support healthcare professionals, patients, and administrators in managing health-related information more effectively. Within this broad domain, PharmaSense specifically addresses the subfields of AI-powered medical data processing, online telemedicine consultation, automated credential verification, and intelligent treatment outcome analysis. Each of these subfields contributes to the platform's core capabilities and together they define the technological and clinical scope of the project.

1.2.1 Healthcare Technology Domain

Healthcare technology encompasses the full range of software systems, hardware devices, data platforms, and digital tools that are used to support, enhance, and deliver medical services and patient management across clinical and non-clinical settings. In the modern healthcare ecosystem, technology serves as both an enabler of efficiency and a driver of improved patient outcomes. Electronic Health Record systems allow doctors to access a patient's complete medical history instantly. Hospital management systems coordinate complex administrative workflows across large institutions. Diagnostic imaging software enables radiologists to examine high-resolution scans with computer-assisted analysis tools. Patient monitoring systems

collect real-time physiological data from wearable sensors and alert clinical staff to changes in a patient's condition.

PharmaSense occupies a specific niche within this broad landscape by focusing on outpatient and remote healthcare delivery. The system helps patients understand their prescriptions without requiring a pharmacist's explanation, enables them to access verified doctors from their home without travelling to a clinic, and gives doctors a structured digital environment in which to provide consultations and manage their patient caseloads. By automating prescription interpretation and facilitating verified remote consultations, PharmaSense reduces the burden on physical healthcare infrastructure while extending the reach of qualified medical care to patients in underserved or geographically remote areas.

1.2.2 Artificial Intelligence in Healthcare

Artificial Intelligence has emerged as one of the most transformative forces in modern medicine. AI systems are now capable of performing tasks that previously required years of clinical training, including analysing medical images for signs of disease, predicting patient outcomes based on complex datasets, identifying drug interactions in electronic prescriptions, and generating personalised treatment recommendations based on a patient's unique medical profile. In the context of

The AI then applies its medical knowledge to generate intelligent predictions about how the patient's treatment is likely to progress under different behavioural scenarios. This use of AI in PharmaSense transforms what would otherwise be a simple document upload into a meaningful clinical support tool that helps patients become active and informed participants in their own healthcare journey. The AI also supports the generation of dietary recommendations and health tips that are specifically tailored to the medicines and conditions identified in the prescription, adding another layer of personalised health guidance.

1.2.3 Medical Data Processing and Analysis

Medical data processing refers to the collection, organisation, transformation, and analysis of health-related information to generate insights that support medical decision-making, patient monitoring, treatment planning, and healthcare administration. In traditional healthcare systems, much of this processing happens manually a doctor reads a patient's file, a pharmacist interprets a handwritten prescription, a nurse records vital signs in a paper chart. These manual processes are time-consuming, prone to human error, and difficult to scale. Digital healthcare platforms like PharmaSense automate and enhance these processes by capturing medical data in structured digital formats, applying computational analysis to extract meaningful patterns, and presenting the results through clear and accessible interfaces.

In PharmaSense, medical data processing begins the moment a patient uploads a prescription document. The system sends the document to the OpenAI API, which performs intelligent extraction of medical entities and returns a structured JSON response containing all identified prescription details. This structured data is then stored in the PostgreSQL database and used to power the treatment analysis and visualisation features. The system also processes consultation records, doctor notes, and uploaded medical documents, maintaining a searchable and organised medical history for each patient that can be accessed by both the patient and their authorised doctors. The use of PostgreSQL's relational data model and Drizzle ORM's type-safe query builder ensures that all medical data is stored consistently, queried efficiently, and protected against structural errors.

1.2.4 Online Healthcare Consultation System

Online healthcare consultation systems are digital platforms that enable remote interaction between patients and medical professionals, overcoming the geographical and logistical barriers that traditionally limited access to medical care. These systems have gained enormous importance in the context of India's healthcare landscape, where a vast population is spread across urban, semi-urban, and rural areas,

and where the doctor-to-patient ratio remains significantly below recommended levels in many regions.

By enabling patients to consult doctors through video calls, text communication, or structured digital request forms, online consultation platforms dramatically expand the reach of qualified medical care. However, a persistent challenge with many online consultation platforms is the lack of robust doctor verification, which creates the risk of patients receiving advice from unqualified or fraudulent practitioners. PharmaSense addresses this challenge directly by integrating automated NMC verification as a mandatory step in the doctor registration process. The NMC (National Medical Commission) is the statutory body established by the Government of India that maintains the official register of qualified medical practitioners in the country. By cross-referencing every doctor's registration number against this official database through the Railway-hosted verification proxy, PharmaSense ensures that only legitimately qualified doctors can provide consultations on the platform. The consultation module also incorporates real-time WebRTC video calls, Stripe-secured payment processing, digital prescription uploads, and comprehensive medical record management, creating a complete and trustworthy telemedicine environment that benefits both patients and doctors.

1.3 OBJECTIVES OF THE STUDY

The primary objective of the PharmaSense project is to design and develop a comprehensive, full-stack, AI-powered telemedicine platform that simultaneously assists patients in understanding their medical prescriptions and connects them with NMC-verified doctors for secure online consultations. This overarching objective encompasses a range of specific technical, functional, and user experience goals that together define the complete scope of the system's intended capabilities.

The first specific objective of the study is to implement an AI-based prescription analysis system that automatically extracts and interprets medical information from uploaded prescription images and PDF documents. This system uses the OpenAI API's multimodal language model, accessed through the Replit AI

Integrations proxy, to perform intelligent optical character recognition and natural language processing on the uploaded prescription content, producing structured medical data without requiring the patient to manually transcribe any information.

The second specific objective is to apply Natural Language Processing techniques through the AI layer to identify, structure, and present medicine details extracted from prescriptions in a clear and patient-readable format. This includes identifying medicine names, dosage values, administration frequency instructions, treatment duration periods, and any disease or condition mentions, and organising these into a structured summary that a patient without medical training can readily understand.

The third specific objective is to integrate an AI-driven treatment outcome forecasting module that generates recovery predictions under three different behavioural scenarios full medication adherence, healthy lifestyle compliance, and non-adherence risk and presents these predictions through interactive visual charts. This feature goes beyond simple prescription interpretation to provide patients with meaningful insights into how their choices and behaviours directly affect their expected health outcomes.

The fourth specific objective is to implement a fully automated NMC doctor verification system using the live Railway proxy that queries the official NMC database to cross-check doctor registration numbers, years of registration, state councils, and supporting personal details before granting a doctor active status on the platform. This verification system is supported by a dual-track approach where successful NMC responses lead to immediate NMC-verified status, and unsuccessful responses due to server unavailability are routed to the administrator for manual review.

The fifth specific objective is to design and implement a secure, role-based authentication and access control system that supports three distinct user roles Patient, Doctor, and Administrator each with appropriate dashboard access, API permissions, and data visibility. This system uses session-based cookie authentication managed by express-session, with all session data stored server-side in the PostgreSQL database to prevent client-side manipulation.

1.4 SCOPE OF THE PROJECT

PharmaSense is scoped as a full-stack web-based telemedicine platform that serves three categories of users patients, doctors, and administrators within the context of the Indian healthcare system and specifically within the regulatory framework of the National Medical Commission. The system is intended to be accessible through any modern web browser on desktop, laptop, and tablet devices, and does not require the installation of any additional software beyond a compatible browser with camera and microphone access for video consultations.

The patient scope of the platform covers the complete patient journey from registration and onboarding through doctor discovery, consultation booking, payment processing, real-time video consultation, prescription receipt, medical record management, and AI-powered prescription analysis. A patient using PharmaSense can create an account with their full personal and health profile, search for doctors using multiple filtering criteria, submit consultation requests with symptom descriptions, complete consultation payments through Stripe, join video calls with their doctor, download prescriptions after the consultation, view their complete medical history, upload their own prescriptions for AI analysis, and track any prescribed medication deliveries through the delivery tracking module.

1.4.1 Functional Scope

The functional scope of PharmaSense covers all features and workflows implemented within the system. The NMC-based automated doctor verification system uses a live Railway proxy with a configurable retry logic that includes a QUICK profile designed for live form lookups with four retry attempts each waiting up to thirty seconds, and a THOROUGH profile designed for registration and re-verification with six retry attempts each waiting up to forty-five seconds. The doctor verification status progresses through a defined sequence of states: pending upon initial registration, nmc_verified when the NMC database confirms the doctor's credentials, manually_approved when the administrator reviews and approves a pending

application, and rejected when the administrator determines that the application cannot be approved.

1.4.2 Data Management and Security Scope

The data management and security scope of PharmaSense covers all aspects of how the system stores, protects, validates, and provides access to medical and personal data. Authentication is implemented using session-based cookies managed by the express-session library, with all session data stored server-side in a dedicated sessions table in the PostgreSQL database. This server-side session storage prevents session data from being tampered with on the client side, as the browser only holds an opaque session identifier cookie and never the actual session payload. Role-based access control is enforced at the API middleware level, where every protected route checks the requesting user's role before processing the request, ensuring that patients cannot access doctor-only endpoints, doctors cannot access admin-only endpoints, and so forth.

1.5 SIGNIFICANCE OF THE PROJECT

The significance of the PharmaSense project extends beyond its technical implementation to address fundamental challenges in Indian healthcare access, prescription safety, and telemedicine trustworthiness. India is home to over 1.4 billion people, yet its healthcare system is characterised by significant disparities in access to quality medical care between urban and rural populations. The doctor-to-patient ratio in many rural and semi-urban areas falls far below the World Health Organisation's recommended threshold, meaning that millions of patients must travel long distances to access even basic medical consultations. PharmaSense directly addresses this accessibility gap by enabling patients to consult verified doctors through their computers or mobile devices without leaving their homes, eliminating both travel time and the risk of exposure to infections in hospital waiting rooms.

CHAPTER 2

LITERATURE SURVEY

2.1 WEB-BASED TELEMEDICINE SYSTEM WITH REAL-TIME COMMUNICATION

A detailed study by Williams and Taylor (2023) in the International Journal of Advanced Research in Computer Engineering presents a web-based telemedicine system that enables real-time communication between doctors and patients. The research focuses on improving remote healthcare services by integrating technologies that support live video and audio consultations, reducing the need for physical visits to hospitals. The authors highlight the importance of timely medical access, especially for patients in remote or underserved areas where healthcare infrastructure is limited and specialist availability is scarce.

The study explains that the proposed system uses web technologies and communication protocols such as WebRTC to facilitate seamless real-time interaction. It includes features such as patient registration, doctor availability status, consultation request handling, and live communication modules. The system is designed to ensure low latency, stable connectivity, and secure data transmission during consultations, providing an effective virtual healthcare experience for both patients and medical professionals. The authors describe the architectural decisions that underpin the platform, including the use of peer-to-peer data channels for media streaming and server-side signalling components that coordinate the establishment of direct connections between participants.

The findings show that real-time telemedicine systems significantly enhance the efficiency of healthcare delivery by enabling immediate medical assistance and reducing waiting times. Patients can consult doctors conveniently from their homes, while doctors can manage multiple consultations effectively.

2.2 SMART HEALTHCARE ASSISTANCE SYSTEM USING ARTIFICIAL INTELLIGENCE

A detailed study by Gupta and Singh (2023) in the Journal of Smart Health Technologies presents an Artificial Intelligence-based healthcare assistance system designed to support medical decision-making and improve patient care. The research focuses on developing an intelligent system capable of analyzing medical data, identifying treatment patterns, and providing healthcare recommendations based on patient information. The authors emphasize that modern healthcare systems generate a large amount of data, and AI technologies can be effectively used to analyze this data and assist healthcare professionals in making informed decisions that are grounded in evidence rather than subjective judgment alone.

The study explains that the proposed system integrates machine learning algorithms with healthcare databases to evaluate patient symptoms, medical history, and treatment records. By analyzing this information, the AI system can provide suggestions related to diagnosis, treatment options, and medication management. The research highlights that such intelligent healthcare assistance systems can improve the efficiency of healthcare services by supporting doctors and medical staff in handling complex medical information. Additionally, the system can help reduce human errors by providing data-driven insights and automated recommendations that flag potential issues before they escalate into clinical complications.

The authors describe the technical implementation of the AI assistance engine in detail, explaining how the system processes both structured data from electronic health records and unstructured data from clinical notes to generate its recommendations. A validation study conducted across multiple healthcare facilities demonstrated that the AI-assisted decision support tool improved diagnostic accuracy and reduced the average time required for clinical assessment.

2.3 INTELLIGENT HEALTHCARE MONITORING SYSTEM USING ARTIFICIAL INTELLIGENCE

A detailed study by Lee and Park (2022) in the IEEE Transactions on Healthcare Systems presents an intelligent healthcare monitoring system that utilizes Artificial Intelligence to monitor patient health conditions and analyze treatment outcomes. The research focuses on developing a smart healthcare framework that continuously collects and analyzes patient data to support effective medical monitoring and decision-making. The authors emphasize that modern healthcare systems require intelligent technologies capable of processing large volumes of patient information and providing accurate insights to healthcare professionals in a timely and actionable format.

The study explains that the proposed system integrates artificial intelligence techniques with healthcare monitoring tools to analyze patient health parameters such as symptoms, treatment responses, and medical history. By applying machine learning algorithms to this data, the system can detect patterns that indicate changes in a patient's health condition and provide alerts or recommendations to healthcare providers. This approach enables continuous monitoring of patients and allows medical professionals to respond quickly to potential health risks or complications before they develop into serious adverse events requiring emergency intervention.

The authors present a detailed evaluation of the monitoring system's performance across a clinical trial involving patients with multiple chronic conditions. The results show that the AI-driven alerting mechanism successfully identified early indicators of health deterioration in a significant proportion of monitored patients, enabling proactive clinical intervention. The study also examines the system's ability to learn from longitudinal patient data, demonstrating that its predictive accuracy improved over time as the model was exposed to a larger and more diverse dataset of patient outcomes and treatment trajectories.

2.4 SECURE HEALTHCARE PLATFORM WITH DOCTOR VERIFICATION

A detailed study by Thomas and Wilson (2022) in the Journal of Digital Health Systems presents a secure healthcare platform that emphasizes doctor verification mechanisms to ensure authenticity and trust in online medical services. The research focuses on addressing the growing concern of unverified medical practitioners in digital healthcare platforms, which can lead to misinformation and reduced patient confidence. The authors highlight the need for reliable verification systems to maintain the integrity of online consultations and protect patients from the risks associated with receiving medical advice from unqualified individuals.

The study explains that the proposed system integrates verification techniques by cross-checking doctor credentials with authorized medical databases and regulatory bodies. It includes features such as secure registration, identity validation, and role-based access control to ensure that only qualified professionals are allowed to provide medical consultations. The system also incorporates data encryption and secure communication protocols to protect sensitive patient information and maintain privacy throughout the consultation process and during the storage and transmission of medical records and prescription data.

The authors conduct a thorough analysis of existing telemedicine platforms and document the verification gaps that exist in many commercially available services. Their research identifies three primary categories of risk arising from inadequate doctor verification: clinical risk stemming from unqualified advice, legal risk arising from unlicensed medical practice, and reputational risk that undermines public trust in digital healthcare as a whole. The proposed verification framework addresses all three categories by implementing a multi-stage credentialing process that validates doctor identities against official medical registration databases before granting access.

2.5 NATURAL LANGUAGE PROCESSING TECHNIQUES FOR HEALTHCARE DATA ANALYSIS

A comprehensive study by Qiu, Zhao, Yang, and Liu (2022) published in IEEE Transactions on Neural Networks and Learning Systems presents a detailed review of Natural Language Processing (NLP) techniques applied to smart healthcare scenarios, covering clinical practice, hospital management, and public health. The research highlights that a large portion of healthcare data exists in unstructured text format, making it difficult for traditional systems to process and interpret medical information effectively. The authors systematically examine NLP approaches including rule-based, statistical, and neural NLP methods, and evaluate their effectiveness in extracting meaningful information from medical documents such as clinical notes, electronic health records, and prescription data.

The study explains that NLP techniques including tokenization, named entity recognition, information extraction, and question-answering systems can identify important medical entities such as drug names, symptoms, dosage instructions, and treatment recommendations, transforming unstructured medical text into structured data suitable for analysis and storage in digital health records. The authors provide a comprehensive taxonomy of NLP tasks in healthcare, categorizing them by application domain and evaluating the state-of-the-art performance of modern transformer-based language models compared to earlier statistical approaches on standard medical NLP benchmarks.

The authors emphasize that NLP-driven healthcare systems can assist doctors, pharmacists, and healthcare administrators by providing faster access to relevant medical information and reducing the burden of manual data interpretation. The review pays particular attention to the role of pre-trained large language models in advancing medical NLP capabilities, noting that models fine-tuned on domain-specific medical corpora consistently outperform general-purpose language models on clinical entity recognition.

2.6 AI-BASED DISEASE PREDICTION USING MEDICAL DATA

A significant study by Lee, Jo, Woo, Im, Park, and Park (2022) published in JMIR AI presents an Artificial Intelligence-based system designed to predict chronic diseases using medical data modeled through a Common Data Model (CDM). The research focuses on how machine learning techniques can be applied to large-scale electronic health record datasets to identify patterns related to disease occurrence and progression. The authors highlight that early detection of chronic diseases such as diabetes, hypertension, hyperlipidemia, and cardiovascular disease is critical for effective treatment, and AI technologies can provide predictive insights based on patient medical records and clinical data.

The study explains that the proposed system applies four validated machine learning algorithms — logistic regression, random forest, gradient boosting, and a neural network classifier — to patient data including medical history, diagnostic reports, and treatment outcomes extracted through the CDM framework. By processing this information, the AI model can detect correlations between health indicators and the likelihood of disease occurrence within a ten-year window. The CDM-based approach ensures that the system can operate across multiple healthcare institutions using different source data formats, making it highly portable and scalable compared to institution-specific prediction models.

The authors conduct a rigorous cross-validation study to assess the predictive performance of each algorithm across all four target conditions, reporting area under the ROC curve scores that demonstrate the practical clinical utility of the AI-driven prediction approach. The study also investigates the interpretability of the prediction models, using feature importance analysis to identify the patient data attributes that contribute most strongly to disease risk predictions. This interpretability analysis is highlighted as a critical component of responsible AI deployment in healthcare, as it enables clinicians to understand and critically evaluate the basis of AI-generated risk assessments.

2.7 MACHINE LEARNING-BASED PRESCRIPTION INTERPRETATION SYSTEM

A detailed study by Ahmed and Rahman (2021) in the International Journal of Machine Learning Applications presents a machine learning-based system designed to automatically interpret medical prescriptions and extract relevant medication information. The research focuses on addressing the challenges associated with understanding handwritten or complex prescription formats, which often lead to misinterpretation of medicine names, dosage instructions, and treatment duration. The proposed system applies machine learning algorithms to analyze prescription data, identify patterns in medical terminology, and accurately detect key treatment details from prescriptions in varying formats and handwriting styles.

By training the model on a dataset of prescription samples, the system learns to recognize common pharmaceutical terms and structures, improving the reliability of automated prescription analysis. The authors describe the data collection methodology in detail, explaining how a diverse corpus of prescription images was assembled from multiple clinical settings to ensure that the model was exposed to the full range of handwriting styles, prescription layouts, and pharmaceutical naming conventions encountered in real-world medical practice. A series of preprocessing steps including image normalization, noise reduction, and binarization was applied to improve the quality of input data before feature extraction.

This structured data can then be used for medication management, patient record maintenance, and treatment monitoring. The research also emphasizes that machine learning techniques can enhance the accuracy of drug identification and dosage recognition, even when prescriptions contain ambiguous handwriting or non-standard formatting. The experimental evaluation demonstrates that the system achieves substantially higher accuracy than rule-based baseline approaches on a held-out test set of prescription images.

2.8 AI-BASED MEDICAL PRESCRIPTION ANALYSIS SYSTEM

A comprehensive study by Kumar and Sharma (2021) in the International Journal of Artificial Intelligence in Healthcare presents an AI-based system designed to automatically analyze medical prescriptions and extract essential treatment information. The research focuses on addressing the challenges associated with interpreting prescription documents, which often contain complex medical terminology and varying prescription formats that are difficult for patients to understand without specialist assistance. The authors propose an artificial intelligence model capable of identifying key components of a prescription, including medicine names, dosage instructions, frequency of administration, and treatment duration.

The study explains that the proposed system utilizes artificial intelligence techniques such as machine learning and pattern recognition to analyze prescription data. By training the system with a dataset of prescription samples, the model learns to recognize medical terms and prescription structures commonly used by healthcare professionals. This enables the system to convert prescription information into structured digital data that can be easily stored, processed, and analyzed. The authors describe the system architecture in detail, outlining how the prescription analysis pipeline is divided into stages covering image preprocessing, text extraction, medical entity recognition, and structured output generation.

The authors highlight that such automation can significantly reduce the workload of pharmacists and healthcare staff who traditionally need to manually review prescription details. The study presents a user evaluation conducted with a sample of patients and healthcare workers, demonstrating that the AI-generated prescription summaries were rated as accurate, clear, and helpful by a large majority of participants. The evaluation also revealed that patients who received AI-generated explanations of their prescriptions reported significantly better understanding of their medication instructions compared to patients who relied solely on the printed prescription and pharmacist.

2.9 DEEP LEARNING APPROACH FOR AUTOMATED PRESCRIPTION RECOGNITION

A detailed study by Patel and Mehta (2021) in the International Journal of Deep Learning Applications presents a deep learning-based system designed for automated recognition of medical prescriptions. The research focuses on addressing the difficulties associated with interpreting handwritten prescriptions, which often contain complex handwriting styles and medical terminology that can lead to misinterpretation. The authors propose a deep learning model capable of identifying and recognizing prescription text, including medicine names, dosage details, and treatment instructions, with improved accuracy compared to traditional recognition methods that rely on fixed rule sets or shallow feature representations.

The study explains that the proposed system utilizes deep learning techniques such as convolutional neural networks and recurrent neural network architectures to analyze prescription images at multiple levels of abstraction. The convolutional layers extract spatial features from the prescription image that capture local handwriting patterns, while the recurrent layers model the sequential dependencies between characters and words in the recognized text. By training the model on a large dataset containing diverse prescription samples collected from multiple hospitals and clinics, the system learns to detect patterns in handwriting and medical terms with a degree of generalization that classical approaches cannot match.

The authors highlight that deep learning models are particularly effective for recognizing complex handwriting patterns that are difficult for conventional Optical Character Recognition (OCR) systems to interpret, especially in cases where characters are poorly formed, ligatures are present, or the prescription uses non-standard abbreviations. A comparative evaluation against three baseline OCR systems demonstrates that the proposed deep learning approach achieves substantially lower character error rates on handwritten prescription samples.

2.10 ONLINE DOCTOR-PATIENT CONSULTATION SYSTEM DESIGN

A detailed study by Singh and Sharma (2021) in the International Journal of Computer Science and Information Technology presents the design and implementation of an online doctor-patient consultation system. The research focuses on improving accessibility to healthcare services by enabling remote interaction between patients and medical professionals through digital platforms. The authors highlight the limitations of traditional healthcare systems, such as long waiting times, geographical barriers, and limited access to specialized doctors, especially in rural areas where the density of qualified medical professionals is significantly lower than in urban centres.

The study explains that the proposed system is developed as a web-based platform that allows patients to register, search for doctors based on specialty and availability, and request consultations either synchronously or asynchronously depending on the urgency of their medical need. Doctors can create detailed professional profiles, manage their availability schedules, and respond to patient consultation requests through a dedicated doctor dashboard. The system incorporates features such as secure user authentication with role-based access control, appointment scheduling with automated confirmation notifications, and basic communication modules to facilitate interaction between doctors and patients within a unified digital environment.

The authors conduct a detailed usability evaluation of the platform with a sample of patients and doctors, collecting feedback on the ease of navigation, the clarity of the booking workflow, and the perceived security of the communication features. The evaluation reveals several areas where the initial design required refinement, particularly in the doctor search interface, where users expressed a preference for more granular filtering options by specialty and geographic region. The study documents the iterative design improvements made in response to this feedback and presents a follow-up evaluation demonstrating measurably improved usability scores after the refinements were implemented.

CHAPTER 3

EXISTING SYSTEM

3.1 MANUAL PRESCRIPTION INTERPRETATION

In the existing healthcare landscape, the process of prescription interpretation remains almost entirely manual and patient-dependent. When a doctor concludes a consultation, they typically provide the patient with a handwritten or printed prescription listing the recommended medications, their doses, and the instructions for taking them. The patient is then expected to read this prescription, understand its contents, and follow the instructions accurately throughout the course of their treatment. In practice, this expectation is frequently unrealistic.

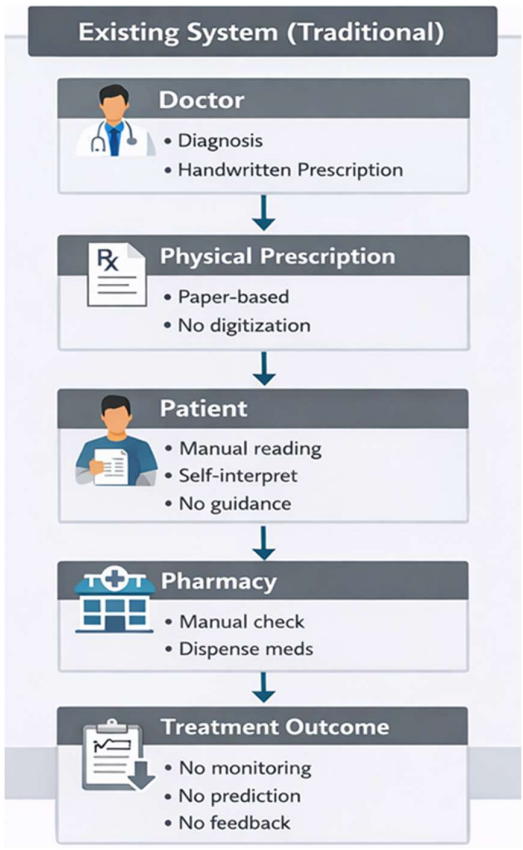

Figure 3.1 Existing System

3.2 LIMITED AUTOMATION IN PRESCRIPTION ANALYSIS

Most existing healthcare applications and digital health platforms focus their automation efforts on the administrative and logistical aspects of healthcare delivery scheduling appointments, storing electronic health records, billing for services, and communicating between healthcare providers rather than on the analytical and interpretive aspects that could most directly benefit patients. Prescription management features in these systems, when they exist at all, typically amount to little more than digital storage of prescription images or typed text. The prescription is stored but not analysed; it is accessible but not explained; it is recorded but not connected to any guidance about how the patient should use it or what to expect from it.

The absence of intelligent prescription analysis in existing systems means that patients who want to understand their prescriptions beyond the basic pharmacist interaction have very limited options. They may search for information about individual medications on the internet, but this approach is risky because general online information about medications is not tailored to the patient's specific dose, condition, or combination of medicines.

3.3 LACK OF INTELLIGENT HEALTHCARE ASSISTANCE

Beyond the immediate interpretation of prescription contents, existing healthcare systems almost universally lack any capacity for intelligent, forward-looking health assistance that could help patients understand how their treatment is likely to progress and what factors might influence their outcomes. A patient who is prescribed a course of antibiotics for a bacterial infection, for example, receives instructions to take the medication at specified intervals for a specified number of days, but receives no automated guidance about what recovery typically looks like under this treatment regimen, how quickly they can expect symptoms to improve, what signs might indicate that the treatment is not working as expected, what dietary or lifestyle practices might support their recovery, or what risks they face if they stop taking the antibiotics before completing the full course.

3.4 LACK OF VERIFIED AND SECURE CONSULTATION

A major and well-documented limitation of existing online healthcare platforms is the inadequacy of doctor verification mechanisms. With the rapid expansion of telemedicine as a convenient and accessible mode of healthcare delivery, there has been a simultaneous rise in the number of fraudulent or unqualified individuals posing as licensed medical practitioners. Many digital health platforms, especially those aiming for quick scalability, often have low barriers to entry and lack strict onboarding procedures. As a result, individuals can create profiles with minimal scrutiny, making it difficult to ensure that all listed practitioners are genuinely certified and experienced. In such environments, patients who seek medical advice online have little to no reliable means of confirming whether the professional they are consulting holds valid medical credentials, is registered with a recognized medical authority, or has relevant clinical expertise. This absence of transparency undermines the credibility of the platform and creates a significant gap in trust between patients and healthcare providers.

The consequences of this lack of secure and verified consultation are serious and far-reaching. Patients may unknowingly receive incorrect diagnoses, inappropriate prescriptions, or misleading health advice from unqualified individuals, which can worsen existing conditions or lead to new health complications. In critical cases, such misinformation can delay proper treatment, resulting in severe health risks or even life-threatening situations. Furthermore, the absence of robust verification mechanisms opens the door for malicious activities such as identity fraud, data misuse, and exploitation of vulnerable patients seeking urgent care. This not only affects individual users but also damages the overall reputation and reliability of digital healthcare systems.

3.5 DRAWBACKS OF THE EXISTING SYSTEM

Despite advances in digital health technology, existing healthcare systems continue to exhibit critical limitations that hinder effective patient care. These drawbacks highlight the urgent need for a more intelligent, integrated, and patient-centric healthcare platform.

- Manual and Error-Prone Prescription Interpretation: Patients are expected to independently read and follow handwritten or printed prescriptions, a process that frequently leads to misinterpretation of medicine names, incorrect dosages, and poor treatment adherence due to complex medical terminology.
- Absence of Intelligent Prescription Analysis: Most existing healthcare applications limit prescription management to digital storage of prescription images or typed text. Prescriptions are stored but never analyzed, leaving patients without any automated explanation or guidance on their treatment.
- No Predictive Health Assistance: Existing systems provide no automated, forward-looking health guidance. Patients receive no information about expected recovery timelines, potential complications, adherence risks, or lifestyle recommendations, leaving them uninformed about their own treatment progress.
- Unverified Doctor Profiles: Many digital health platforms have low barriers to entry and lack strict doctor onboarding procedures, allowing unqualified individuals to pose as licensed practitioners. This absence of verification creates serious risks of misdiagnosis and inappropriate prescriptions.
- Lack of Real-Time Consultation: Existing systems rarely offer integrated real-time video consultation. Patients are typically redirected to third-party tools for communication, breaking the continuity of the digital healthcare experience and introducing security risks.
- Fragmented Healthcare Services: Current platforms address only isolated aspects of healthcare delivery such as appointment scheduling, billing, or record storage,

without providing a unified workflow that connects all stages of a patient's healthcare journey from consultation to prescription management.

3.6 NEED FOR THE PROPOSED SYSTEM

The limitations identified in the existing healthcare systems collectively demonstrate a clear and pressing need for an intelligent, integrated, and patient-empowering digital healthcare platform. The PharmaSense system is proposed as a direct response to these gaps, driven by the growing demand for accessible, accurate, and trustworthy healthcare services in the digital age.

Patients today require more than a passive digital record of their prescriptions. They need to understand what their medications do, how to take them correctly, and what health outcomes they can realistically expect. At the same time, the expansion of telemedicine has made it increasingly important for online platforms to guarantee the authenticity of the medical professionals they list, and to provide real-time communication tools that match the quality of in-person consultations. The PharmaSense system addresses all of these needs simultaneously.

By integrating AI-based prescription analysis, predictive treatment outcome modeling, verified doctor access through the NMC database, and real-time WebRTC video consultation into a single cohesive platform, PharmaSense eliminates the fragmentation and inadequacy of existing solutions. The need for such a system is not only justified by the drawbacks of the current landscape but also supported by the growing availability of AI and web technologies capable of delivering these capabilities in a secure, efficient, and user-friendly manner.

CHAPTER 4

PROBLEMS IDENTIFIED

The development of PharmaSense was motivated by a detailed analysis of the problems and limitations that affect both patients and healthcare providers in the current healthcare technology landscape. Each of the problems identified below represents a specific gap or failure mode in existing systems that the PharmaSense platform is designed to address through its combination of AI, automation, and secure digital infrastructure.

The second identified problem is the complete absence of automated prescription analysis tools in most existing healthcare platforms and patient-facing applications. Despite the critical importance of prescription comprehension to patient safety and treatment outcomes, existing systems rarely provide any feature that analyses the content of a prescription and presents the information in a patient-friendly format. Patients who want an explanation of their prescription beyond what they can obtain from a busy pharmacist have no reliable digital resource to turn to.

The fourth problem is medication non-adherence, which is both a cause and a consequence of the problems described above. When patients do not understand their prescriptions, do not appreciate the importance of completing their treatment, or receive no reminders or prompts to take their medications consistently, they are much more likely to skip doses, take medications irregularly, or stop treatment as soon as they begin to feel better rather than completing the full prescribed course. Medication non-adherence is estimated to be responsible for a significant proportion of treatment failures and hospital readmissions in both developed and developing countries.

The fifth identified problem is the absence of medication alerts and warnings in patient-facing digital health tools. While many healthcare applications include medication reminder features of varying sophistication, very few provide intelligent warnings that alert patients to the clinical consequences of missed doses or the risks associated with specific medication interactions or non-adherence patterns. Without

such warnings, patients who miss doses may not appreciate that this could significantly affect their treatment outcomes.

The sixth problem is the limited use of advanced technologies such as AI, large language models, and multimodal processing in current prescription management systems. Despite the proven capability of these technologies to extract and interpret medical information from unstructured documents, their application to patient-facing prescription management remains largely limited to research and pilot projects rather than deployed consumer-facing products.

The seventh problem relates to the inefficient utilisation of the rich medical data contained in prescription documents. A prescription is not merely a list of instructions; it is a structured clinical document that, when properly analysed, reveals information about the patient's diagnosed condition, the doctor's assessment of the severity of that condition, the expected duration of the illness, and the therapeutic strategy chosen to address it. Existing systems treat this data as opaque and leave it unanalysed when, with appropriate AI tools, it could be converted into valuable patient guidance.

The eighth problem is the absence of reliable doctor verification in most online healthcare consultation platforms. As described in the existing system analysis above, the lack of programmatic cross-referencing against official medical registries means that patients on most platforms cannot reliably verify whether their consulting doctor holds the qualifications they claim.

The ninth problem is the difficulty patients face in finding doctors who are suitable for their specific needs. A patient who requires a specialist in a particular medical field, who prefers to consult in a specific regional language, or who lives in a particular geographic area may struggle to find a suitable doctor through existing platforms that lack intelligent filtering and matching capabilities.

CHAPTER 5

PROPOSED SYSTEM

5.1 AI-BASED PRESCRIPTION ANALYSIS SYSTEM

The proposed PharmaSense system addresses the first and most critical gap in existing healthcare applications by implementing a sophisticated AI-based prescription analysis feature that transforms the patient's experience of receiving and understanding a medical prescription. Rather than leaving patients to struggle with medical terminology and pharmacist-dependent interpretation, PharmaSense provides an automated, intelligent, and highly accessible prescription analysis tool that any patient can use from their home simply by uploading a photograph or digital copy of their prescription.

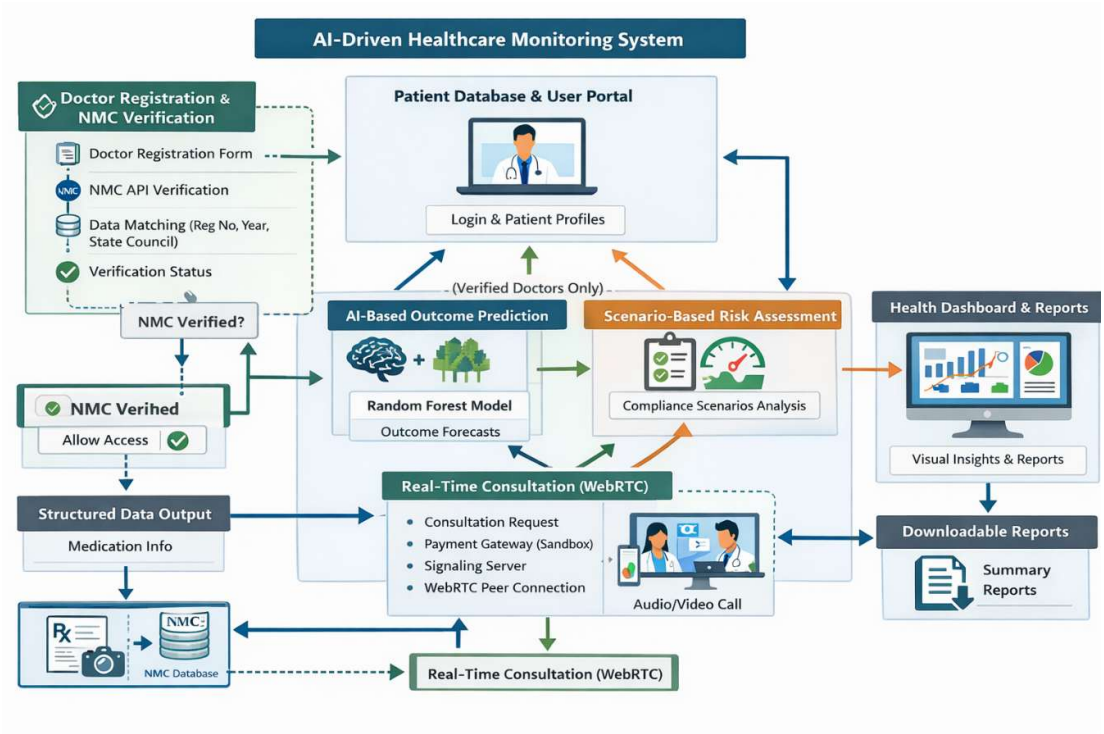

Figure 5.1 Proposed System

5.2 PREDICTIVE TREATMENT ANALYSIS AND HEALTH INSIGHTS

PharmaSense advances significantly beyond the capabilities of a simple prescription reader by incorporating an AI-driven treatment outcome prediction engine that generates forward-looking health insights tailored to the specific medications and conditions identified in the patient's prescription. This predictive analysis module represents one of the most innovative and patient-empowering features of the platform, providing guidance that no existing patient-facing digital health application currently offers in a form accessible to ordinary users.

The predictive analysis works by combining the structured prescription data extracted in the analysis stage with contextual inputs about the patient's current treatment progress. The patient is asked to indicate how many days of their treatment they have completed so far, and this information is combined with the prescribed treatment duration extracted from the prescription to calculate the remaining treatment period. The model predicts the expected recovery timeline when the patient takes every dose at the prescribed time and completes the full course of treatment.

5.3 VISUAL HEALTH REPORTING AND INTELLIGENT HEALTHCARE ASSISTANCE

In addition to providing structured prescription data and narrative health guidance, PharmaSense presents treatment analysis results through interactive visual charts that allow patients to see and compare their different recovery scenarios at a glance. The frontend interface uses the Recharts library, specifically the ComposedChart component with Area and Line sub-components, to create visually rich and interactive graphs that plot the expected recovery progress over time for each of the three prediction scenarios. The x-axis of these charts represents time in days from the start of the treatment, and the y-axis represents a normalised recovery progress indicator that goes from zero at the start of treatment to one hundred at full recovery.

The visualisation system is also used in the administrator analytics dashboard, where Recharts ComposedChart components display platform-wide metrics

including user registration trends over time, consultation volume by week or month, and revenue summaries. The administrator dashboard provides these analytics through a purpose-built API endpoint that aggregates data from the consultations, users, and payments tables in the PostgreSQL database and returns them in a format optimised for Recharts rendering. All gradient definitions in the charts use unique IDs to prevent CSS conflicts between multiple charts displayed on the same page, a technical detail that ensures visual consistency across all chart instances.

5.4 REAL-TIME CONSULTATION AND DIGITAL HEALTH MANAGEMENT

The Doctor–Patient Consultation Platform addresses the problems of unverified doctors, inaccessible specialists, and limited real-time consultation through a comprehensive digital workflow that takes a patient from doctor discovery through verified consultation to digital prescription receipt in a single integrated platform experience. The patient can filter these results by medical specialty to find a doctor whose area of expertise matches their health concern, by preferred language to ensure the consultation can be conducted in a language they are comfortable with, and by geographic region to find doctors who have familiarity with their local healthcare context.

The WebRTC video consultation uses STUN server configuration to traverse NAT and firewall boundaries that would otherwise prevent a direct peer-to-peer connection between the patient's and doctor's browser instances. The API server acts as a signalling server to coordinate the WebRTC handshake by relaying the offer, answer, and ICE candidate messages between the two parties until a direct data channel is established.The video interface provides controls for muting the microphone, disabling the camera, and ending the call, and displays both the local stream in a picture-in-picture thumbnail and the remote stream in the main view area. After the consultation concludes, the doctor can upload the patient's prescription as a PDF or image file through the prescription upload interface.

5.5 ADVANTAGES OF THE PROPOSED SYSTEM

The PharmaSense system introduces a range of significant advantages over conventional healthcare applications by combining AI-based prescription analysis, predictive health insights, real-time video consultation, and verified doctor access into a single integrated platform. These advantages directly address the gaps identified in the existing system and substantially improve the overall healthcare experience for patients, doctors, and administrators.

- Automated Prescription Analysis: PharmaSense enables patients to upload a photograph or digital copy of their prescription and receive an instant AI-driven breakdown of medication names, dosage instructions, frequency, and treatment duration, eliminating the need for manual interpretation or pharmacist dependency.
- Predictive Treatment Insights: The platform provides forward-looking recovery timelines and outcome predictions based on the patient's current treatment progress, offering guidance that no existing patient-facing healthcare application currently provides in an accessible format.
- Interactive Visual Health Reports: Treatment outcomes and recovery scenarios are presented through interactive Recharts-powered graphs, allowing patients to visually compare expected recovery progress under different adherence conditions at a glance.
- Verified Doctor Access: All doctors on the platform are verified against the National Medical Commission registration database through the NMC verification proxy, ensuring that patients consult only qualified and licensed medical professionals.
- Real-Time WebRTC Video Consultation: Patients can conduct live video consultations with doctors directly through the browser using WebRTC, removing geographical barriers and eliminating the need for third-party communication tools.

CHAPTER 6

SYSTEM REQUIREMENTS

System requirements describe the hardware and software components necessary to develop and run the PharmaSense system effectively. These requirements ensure that the system performs critical functions such as prescription image processing, AI-based analysis, real-time video consultations, and medical records management without performance issues. The system requirements are categorized into hardware requirements and software requirements.

6.1 HARDWARE REQUIREMENTS

Hardware requirements refer to the physical components needed to run the PharmaSense system. The system is designed as a web-based application and does not require specialised or expensive hardware. A standard modern computer with adequate processing capability is sufficient to handle prescription uploads, WebRTC video consultations, and AI-based analysis smoothly.

- **Processor:** Intel Core i3 or equivalent (i5 or equivalent recommended for frequent video consultations)
- **RAM:** Minimum 4 GB (8 GB recommended for better performance)
- **Storage:** Minimum 500 GB Hard Disk or SSD
- **Internet Connection:** Minimum 2 Mbps for general use; 10 Mbps recommended for WebRTC video consultations
- **Camera and Microphone:** Minimum 720p camera required for video consultation participation
- **Supported Processors:** x86-64 (Intel/AMD) and ARM-based processors (modern tablets and laptops)

6.2 SOFTWARE REQUIREMENTS

Software requirements refer to the programs, platforms, and development tools used to build and operate the PharmaSense system. The system is structured as a pnpm monorepo with two primary artifacts: an API server and a web frontend, supported by a range of frameworks, libraries, and external service integrations.

- **Operating System:** Windows 10 / 11, Linux (Ubuntu 20.04 or later), or macOS 12 or later
- **Runtime Environment:** Node.js version 20 LTS with pnpm package manager
- **Frontend Technologies:** React 18, TypeScript, Vite, Tailwind CSS v4, Recharts, Framer Motion, Lucide React
- **Backend Framework:** Node.js with Express and TypeScript; express-session for authentication; Multer for file uploads; pino for structured logging
- **Database:** PostgreSQL with Drizzle ORM for type-safe, ACID-compliant data management
- **AI Integration:** OpenAI GPT-4o (via Replit AI proxy) for prescription analysis and treatment outcome prediction
- **Payment Integration:** Stripe (test mode) for consultation payment processing
- **Video Consultation:** WebRTC with STUN server configuration for peer-to-peer real-time video
- **Development Tools:** TypeScript, ESBuild, dotenv; all secrets managed via platform environment variables

These software components collectively enable the PharmaSense system to support prescription analysis, secure medical records management.

CHAPTER 7

SYSTEM IMPLEMENTATION

7.1 LIST OF MODULES

1. Authentication, Role Management
2. Doctor Registration, NMC Verification
3. Consultation Request, Payment And Appointment
4. Real-Time Video Consultation
5. AI Prescription Analyser

7.2 MODULE DESCRIPTION

MODULE 1: AUTHENTICATION, ROLE MANAGEMENT

The Authentication, Role Management and Admin Dashboard Module forms the combined security foundation and administrative oversight layer of the entire PharmaSense platform, providing both the mechanisms through which users establish their identity and maintain persistent authenticated sessions, and the tools through which the platform administrator governs the health and integrity of the system. The module supports three distinct user roles Patient, Doctor, and Administrator each of which has a separate registration pathway, a dedicated dashboard interface, and a specific set of API-level permissions enforced at the server level. This role boundary is not a frontend convention but a server-enforced contract: every protected route checks the requesting user's session role before processing the request, ensuring that patients cannot access doctor-only endpoints, doctors cannot access admin-only endpoints, and administrators retain exclusive access to all platform governance functions.

The registration process for patients and doctors is handled through dedicated registration forms on the frontend that collect all required information for the respective

user type. Patient registration collects personal details including full name, email address, password, date of birth, blood group, known allergies, and existing medical conditions. Doctor registration collects the same core account details plus NMC registration number, year of NMC registration, state medical council, medical specialty, languages spoken, geographic region, and consultation fee.

The administrator account is pre-configured in the system with the email address admin@pharmasense.in, and access to all administrator features is restricted exclusively to users who have authenticated with this account. The administrator dashboard is inaccessible to patients and doctors through both frontend routing guards and API-level role checks. The most critical administrative function is the review and approval of pending doctor registrations.

The administrator analytics dashboard provides a visual overview of platform performance metrics including total registered patients, total registered doctors broken down by verification status, total completed consultations, and cumulative revenue from consultation payments.

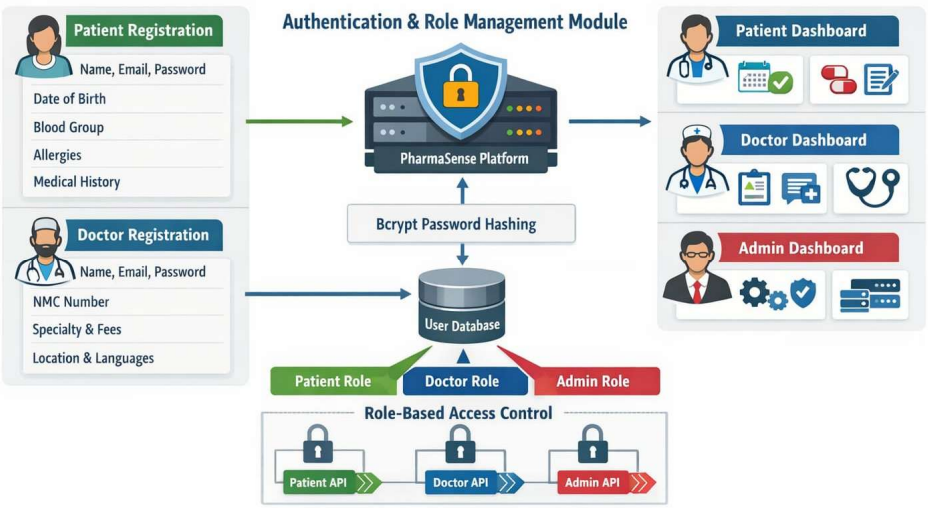

Figure 7.1 Authentication, Role Management

MODULE 2: DOCTOR REGISTRATION, NMC VERIFICATION

The Doctor Registration, NMC Verification and Search Module manages the complex multi-step process of onboarding new doctors onto the PharmaSense platform including the critical NMC credential verification step that distinguishes PharmaSense from less rigorous online consultation platforms as well as the mechanism through which patients discover and select those doctors for consultations. Together, these two closely related functions form the complete doctor supply side of the platform: doctors are verified and approved, and patients are given the tools to find and select among them. This module represents one of the most technically demanding components of the system due to the challenges of integrating with the NMC government database infrastructure, which is characterised by variable response times and inconsistent availability.

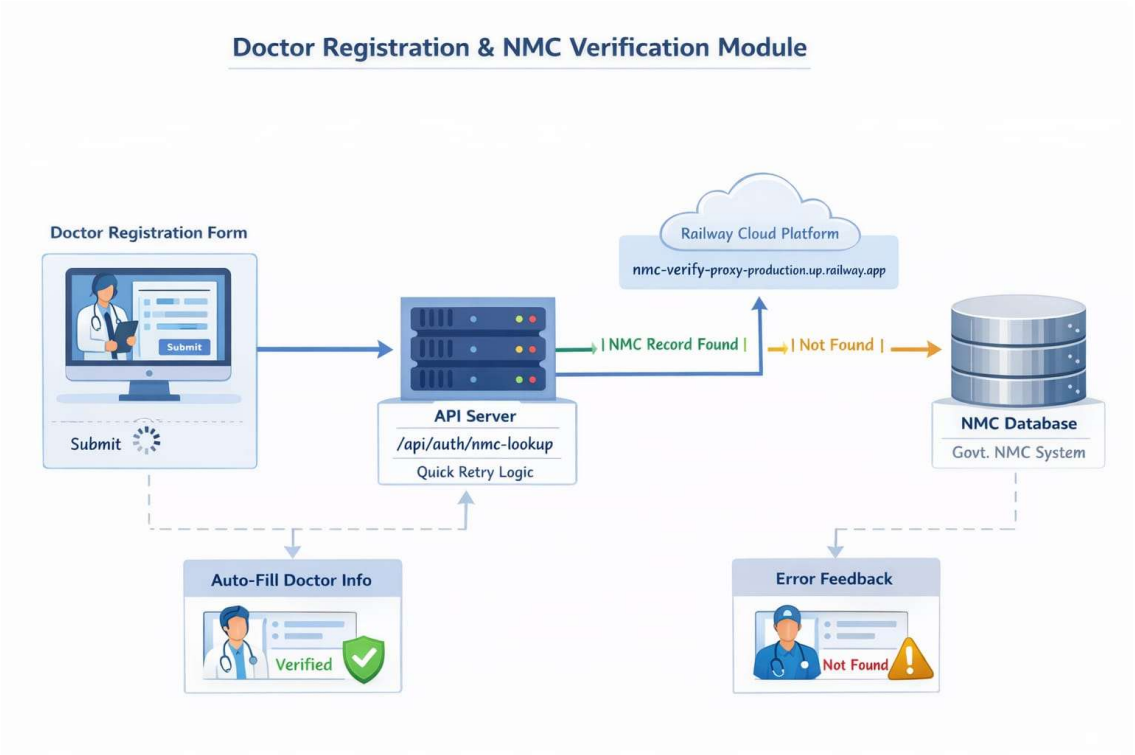

Figure 7.2 Doctor Registration, NMC Verification

This lookup uses the QUICK retry profile, which makes up to four attempts to reach the NMC verification proxy with a thirty-second timeout per attempt. The NMC verification proxy is a separate service hosted on the Railway cloud platform at the URL https://nmc-verify-proxy-production.up.railway.app/api/nmc-verify, which accepts a registration number as a query parameter and returns the corresponding NMC record if found.

The Doctor Search and Filtering capability of this module provides the primary mechanism through which patients discover suitable doctors for their consultations. The search interface presents patients with a set of filtering controls that allow them to specify their requirements across multiple dimensions simultaneously. A specialty filter allows the patient to select from all medical specialties represented by active doctors on the platform, including General Medicine, Paediatrics, Dermatology, Cardiology, Orthopaedics, Gynaecology, Psychiatry, Ophthalmology, ENT, and others depending on the doctors registered at any given time.

MODULE 3: CONSULTATION REQUEST, PAYMENT AND APPOINTMENT

The Consultation Request, Payment and Appointment Scheduling Module manages the complete lifecycle of a consultation transaction from the patient's initial request through payment confirmation and status advancement to the video consultation stage, as well as providing both patients and doctors with a structured, calendar-aware view of their upcoming and historical appointments. This module integrates Stripe's payment processing service as an external dependency into the consultation workflow, creating a coherent and fully trackable transaction record that drives the status progression of every consultation on the platform.

When a patient submits a consultation request, the frontend collects the patient's symptom description, any relevant medical history notes they wish to share with the doctor, and their preferred consultation time. This information is submitted to the /api/consultations endpoint, which creates a new consultation record in the database with a status of pending. The consultation record links the patient and doctor by their respective database identifiers and stores all submitted request information alongside

the target consultation date and time. The doctor is notified of the new request through their dashboard, which displays all pending consultation requests with the patient's full submitted information. The doctor reviews the incoming request, reads the symptom description, and decides whether to accept or decline. Upon acceptance, the consultation status is updated to accepted and the patient is notified that they may proceed to payment.

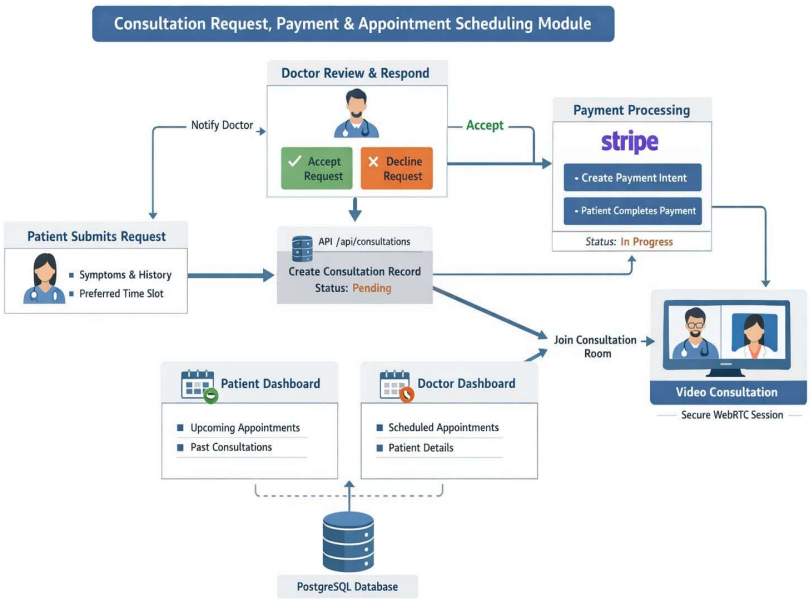

Figure 7.3 Consultation Scheduling

For doctors, the appointments view provides a schedule management tool that shows all upcoming consultations with patient details and the scheduled time, allowing the doctor to plan their working day and ensure they are appropriately prepared for each patient's specific concern. The scheduling data is drawn directly from the consultations table in the PostgreSQL database and updated in real-time as consultation statuses change. The scheduling interface uses the consultation record's scheduled time field and status field to determine which appointments should appear in upcoming, in-progress, or completed categories, and renders them in the appropriate section of the interface accordingly.

MODULE 4: REAL-TIME VIDEO CONSULTATION

The Real-Time Video Consultation, Prescription Management and Health Records Module provides the technical infrastructure for live peer-to-peer consultation, the complete lifecycle management of prescription and medical document files, and the centralised patient health record system that accumulates and organises all medical information generated through the platform. These three closely related capabilities form the clinical core of the PharmaSense platform: the video call is where the consultation happens, the prescription is its primary output, and the health records system ensures that both persist in a form accessible and useful beyond the immediate appointment.

Both parties then exchange ICE candidates network address and port combinations that can potentially be used to establish a direct peer-to-peer connection through the same server-side signalling relay. The STUN server provides information

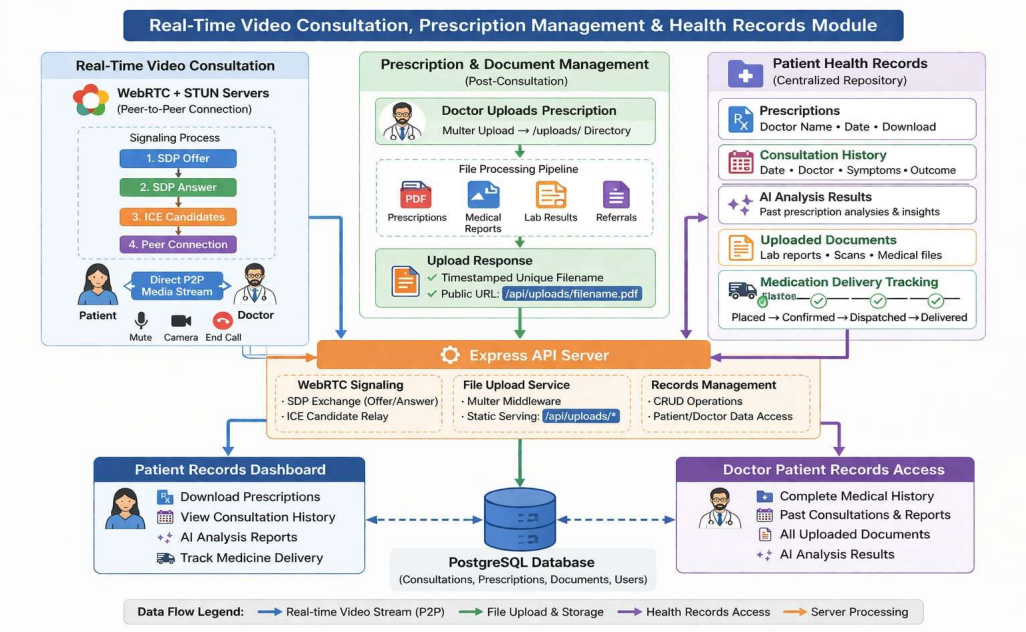

Figure 7.4 Real-Time Video Consultation

about the public IP addresses that each browser is reachable at from outside their local network, which is essential for establishing a direct connection when one or both parties are behind NAT. Once ICE negotiation completes and a suitable connection path is found, the peer connection is established and all subsequent audio and video data flows

directly between the two browsers without passing through the server, ensuring minimal latency and optimal call quality. The consultation interface displays the remote stream in the main view area and the local stream in a picture-in-picture overlay. Both participants have controls to mute the microphone, disable the camera, and end the call, and the interface displays connection status feedback during establishment.

AI prescription analysis results are stored and accessible in a dedicated section of the records view, allowing patients to review past analyses and track how their treatment understanding has evolved over time. For doctors, the health records component provides access to the complete medical history of any patient they are consulting with or have previously seen, giving them the context needed for informed clinical decisions during new consultations without relying on the patient's verbal recall. The Medication Delivery Tracking capability further enriches the records view by providing patients with a status timeline for any prescribed medication orders placed through the platform, progressing through defined states of order placed, confirmed by the pharmacy, dispatched, and delivered, giving patients a clear and reassuring view of when they can expect to receive their medications following a consultation.

MODULE 5: AI PRESCRIPTION ANALYSER

The AI Prescription Analyser Module is the most technologically sophisticated component of the PharmaSense platform and represents the clearest demonstration of how advanced artificial intelligence can be applied to improve patient understanding and active engagement with their own healthcare. This module is accessible from the patient dashboard through a dedicated navigation section and provides a standalone prescription analysis experience that functions completely independently of any active consultation on the platform, meaning patients can use it to analyse prescriptions received from any source not only those issued through PharmaSense itself.

The patient navigates to the AI Prescription Analyser, uploads their prescription document either as an image file in JPG or PNG format or as a PDF document, and optionally provides additional context such as the number of days they have already

been taking their medication. The uploaded file is processed by Multer on the API server and stored on the filesystem in the uploads directory. The server then reads the file from disk and encodes it as a Base64 string, which is used to construct a multimodal data URI of the form followed by the encoded content. This data URI is included in a structured request sent to the OpenAI API through the Replit AI Integrations proxy, which routes the request to the OpenAI endpoint and handles API key management transparently. The request also includes a carefully engineered natural language prompt that instructs the GPT-4o multimodal language model to perform a complete prescription analysis and return results in a structured JSON format.

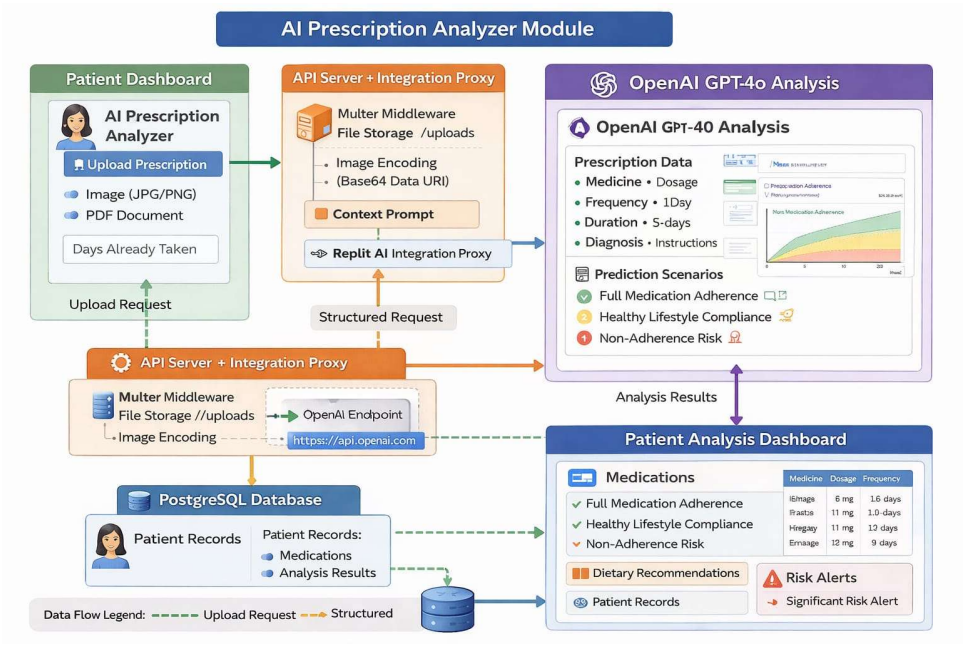

Figure 7.5 AI Prescription Analyser

The OpenAI GPT-4o model processes both the visual content of the prescription image and the text prompt simultaneously, leveraging its extensive training on medical and pharmaceutical literature to extract structured data from the prescription document regardless of whether it is handwritten, printed, or in a mixed format. The model identifies and extracts all medicine-related entities from the prescription, including the generic and brand names of each medication, the prescribed dosage in appropriate units, the frequency of administration expressed in comprehensible terms such as once daily or twice daily rather than the abbreviated Latin notations that appear in the original

prescription, the duration of the treatment course, and any associated medical condition or diagnosis mentioned in the document.

The model also identifies any special instructions associated with specific medications, such as whether they should be taken with food, at specific times of day, or with a particular fluid quantity.The model's complete response is returned as a structured JSON object containing an array of identified medications with all their extracted details, the three scenario prediction objects with timeline data and risk descriptions, and a list of dietary and lifestyle guidance strings.

7.3 SYSTEM ARCHITECTURE

PharmaSense is implemented as a pnpm monorepo containing two primary application artifacts. The API server artifact is a Node.js Express TypeScript application that runs on port 8080 in the Replit environment. The mediconnect web frontend artifact is a React Vite TypeScript application that is served by the Vite development server during development and compiled to a static bundle for production deployment. The two artifacts communicate through a defined REST API contract where the frontend makes HTTP requests to the API server through the /api/ path prefix, which the Replit reverse proxy routes to the API server. The API server's routes are organised into dedicated route files for authentication, doctor management, patient management, consultation management, payment processing, file upload, administrator functions, and NMC verification lookups.

The database layer consists of a PostgreSQL database with a schema defined in TypeScript using the Drizzle ORM. The schema includes tables for users, doctors, patients, consultations, prescriptions, medical documents, appointments, and sessions. All table definitions use serial integer primary keys. Foreign key relationships are defined between the users table and both the doctors and patients tables and between the medical documents table and the patients table.

Table 7.1 Treatment Analysis and Prediction

Type of Analysis	Description	Example Scenario
Prescription Text Extraction Analysis (OCR Processing)	Extracts textual information from uploaded prescription images using Optical Character Recognition (OCR). The system converts handwritten or printed prescriptions into machine-readable text that can be processed for further analysis.	A patient uploads a prescription image containing "Paracetamol 500mg – twice daily for 5 days". The OCR module extracts the text and converts it into digital form for further processing.
Medicine Information Structuring (NLP Processing)	Uses Natural Language Processing (NLP) techniques to identify and structure medical entities such as medicine name, dosage, intake frequency, and duration. The extracted information is standardized and prepared for predictive analysis.	From the extracted text, the system identifies the medicine name "Paracetamol", dosage "500 mg", frequency "2 times/day", and duration "5 days", and converts them into structured fields for prediction models.
Treatment Outcome Forecasting Analysis	Applies Random Forest machine learning algorithms to analyze prescription data, patient profile, and lifestyle inputs to forecast recovery duration, symptom improvement, and potential risk levels under different adherence scenarios.	For a patient diagnosed with mild fever and prescribed Paracetamol for 6 days, the system predicts that with proper adherence recovery may occur within 5–6 days with a 92% success probability.

Table 7.2 Online Consultation System Functional Analysis

Type of Analysis	Description	Example Scenario
Doctor Verification Analysis	Validates doctor credentials by comparing registration details with the NMC database. Ensures authenticity before allowing access to the platform.	A doctor enters registration number, year, and state council. The system matches these with NMC records and verifies the doctor automatically if all details are correct.
Doctor Search and Filtering Analysis	Analyzes patient input such as symptoms, location, and preferences to recommend suitable doctors using intelligent filtering.	A patient enters symptoms like “skin rash” and selects Tamil language. The system suggests dermatologists available online in the selected region.
Consultation and Communication Analysis	Manages the consultation workflow including request handling, response from doctors, and real-time communication via video/audio calls.	A patient sends a consultation request. The doctor accepts it, and the system enables a video call session after payment.
Medical Record and Prescription Management Analysis	Stores and organizes patient medical history, prescriptions, and consultation data for easy access and future reference.	After consultation, the doctor uploads a prescription. The patient can view and download it later from their dashboard.

CHAPTER 8

SYSTEM TESTING

8.1 UNIT TESTING

Unit testing is the process of testing individual components, functions, and modules of the system in isolation to verify that each unit of code behaves correctly given a range of inputs. In the PharmaSense system, unit testing was performed across both the frontend and backend during the development process to verify the correctness of individual components before integration. On the backend, unit testing focused on the NMC verification functions, the authentication middleware functions, the file upload processing logic, and the AI prescription analysis request construction and response parsing logic.

The NMC verification module was tested with a range of registration numbers, including known valid numbers from the NMC database, numbers that are not in the NMC database, and requests made when the NMC proxy is simulated to be unavailable, to verify that all three cases are handled correctly and that the appropriate status is returned in each scenario. The authentication middleware was tested to verify that it correctly identifies authenticated sessions, correctly identifies the user's role from the session data, and correctly rejects requests without valid sessions or with insufficient role permissions. On the frontend, unit testing focused on the form validation logic in the registration forms, the NMC status display logic in the doctor registration form, and the chart rendering logic in the AI analysis results view.

8.2 INTEGRATION TESTING

Integration testing verifies that individual modules work correctly when combined and that data flows correctly through the system from one module to another. In the PharmaSense system, integration testing focused primarily on the end-to-end data flows that cross module boundaries, particularly the registration-to-verification-to-approval flow for doctor onboarding, the search-to-request-to-payment-to-consultation

flow for patient consultations, and the upload-to-storage-to-download flow for prescription document management.

The registration-to-verification-to-approval flow was tested by registering new doctor accounts through the frontend registration form, observing the NMC verification attempt in the API server logs, confirming that the correct verification status was applied to the doctor record in the database, and then approving the doctor through the administrator dashboard to verify that the status advanced correctly and the doctor became visible in patient search results.

The search-to-consultation flow was tested by performing a doctor search from a patient account, submitting a consultation request, accepting the request from the doctor's dashboard, completing a Stripe test payment using a Stripe test card number, and verifying that the consultation status advanced to in-progress and that the video consultation room became accessible to both parties. The upload-to-download flow was tested by uploading a PDF prescription from a doctor account, copying the generated download URL from the prescription record, and verifying that requesting that URL returned the correct PDF file with the correct content type header rather than the HTML page that would result from incorrect routing.

8.3 SYSTEM TESTING

System testing evaluates the complete PharmaSense application as an integrated whole, verifying that all modules work together correctly to support the defined user journeys from end to end. System testing covered the complete patient journey from account creation through doctor search, consultation booking, payment, video consultation, prescription receipt, and AI analysis. It also covered the complete doctor journey from registration through NMC verification, administrator approval, consultation management, and prescription upload. The administrator journey was tested to verify that all pending doctor applications were visible, that approval and rejection actions correctly updated doctor statuses, and that all platform analytics were accurately calculated and displayed.

System testing also evaluated the error handling behaviour of the system under exceptional conditions, including attempting to register with an email address that is already in use, attempting to submit a consultation request to a doctor who is not approved, attempting to access a video consultation room without completing payment, and attempting to access administrator features from a patient or doctor account.

8.4 PERFORMANCE TESTING

Performance testing evaluated the responsiveness and efficiency of the PharmaSense API server and frontend under realistic usage conditions. The API server was tested for response times across all major endpoint categories, including authentication endpoints, doctor search queries with multiple filter parameters, consultation status updates, and prescription upload operations. The AI prescription analysis endpoint was specifically evaluated for response time given the external API call to OpenAI, with typical response times of fifteen to thirty seconds for prescription processing confirmed as acceptable given the complexity of the analysis performed.

The NMC verification system was specifically performance-tested given the known variability of the NMC government server response times. Tests confirmed that the QUICK retry profile successfully retrieves NMC records when the government server responds within thirty seconds per attempt, and that the THOROUGH profile provides additional retry capacity for the registration verification path. When the NMC server returns HTTP 500 errors from the Railway proxy due to its own fifteen-second timeout, the retry logic correctly cycles through the configured number of attempts before gracefully falling back to the pending status with appropriate logging.

8.5 SECURITY TESTING

Security testing verified that the PharmaSense system correctly implements all defined security controls and that no obvious vulnerabilities exist in the authentication, authorisation, or data access control mechanisms. Password storage was verified to use bcrypt hashing with an appropriate cost factor, confirming that plaintext passwords are never stored in the database. Session cookie security was verified to confirm that

cookies are set with the httpOnly attribute, preventing JavaScript access from browser-side scripts. Role-based access control was verified by attempting to access endpoints designated for one role from an account of a different role, confirming that all cross-role access attempts are rejected with appropriate HTTP 401 or 403 responses.

File upload security was verified by attempting to upload files with disallowed content types and by verifying that the Multer configuration correctly rejects non-image and non-PDF uploads. The /api/uploads/ routing was verified to confirm that uploaded files are correctly served through the Express static middleware and that the correct MIME types are included in the Content-Type response header for different file types.

8.6 USABILITY TESTING

Usability testing evaluated the ease and intuitiveness of the PharmaSense user interface for users with varying levels of technical and medical familiarity. The doctor registration form was specifically evaluated for its handling of the NMC verification process, which is technically complex but must be presented to doctors in a simple and reassuring way.

The loading banner that appears during the NMC verification process, which informs the doctor that verification may take up to sixty seconds on the government server, was found to effectively manage doctor expectations and reduce anxiety about the registration process. The dark medical theme was evaluated for readability and contrast across all pages, with particular attention to the opacity-based colour modifiers used for status indicators and alert banners. The AI prescription analysis results page was evaluated to confirm that the information is presented in a way that is comprehensible to patients without medical training.

CHAPTER 9

RESULT AND DISCUSSION

The PharmaSense system was developed as a full-stack telemedicine platform integrating AI prescription analysis, NMC-based doctor verification, real-time WebRTC video consultation, and comprehensive medical records management into a single unified application. Following the completion of the implementation and the execution of the testing phases described in the previous chapter, a range of results were observed that confirm the system's functional capabilities and highlight areas of continued development.

The authentication and role management system performed as expected across all tested scenarios. Patient registration, doctor registration, and administrator login all functioned correctly, with appropriate session cookie issuance on successful authentication and role-based access control correctly enforced across all protected API endpoints. The database transaction wrapping for registration operations successfully prevented partial records in all tested failure scenarios, and the HTTP 409 response for duplicate email registrations provided clear and actionable feedback to users who attempted to register with an already-used email address.

The NMC verification system demonstrated the expected pattern of behaviour given the characteristics of the NMC government server infrastructure. In testing with known valid registration numbers such as 146631, the NMC proxy successfully returned the doctor's record when the government server responded within the Railway proxy's fifteen-second window, achieving response times as low as 14.8 seconds in successful cases. In the majority of cases with other registration numbers, the NMC government server consistently took longer than fifteen seconds to respond, causing the Railway proxy to return HTTP 500 errors that triggered the retry logic and ultimately led to the pending fallback status.

CHAPTER 10

CONCLUSION AND FUTURE WORK

10.1 CONCLUSION

The PharmaSense system has been successfully developed and tested as a comprehensive, AI-powered, full-stack telemedicine platform that addresses a carefully researched set of real-world problems in patient prescription understanding, online healthcare accessibility, and doctor credential verification in the Indian healthcare context. The platform brings together a diverse range of modern technologies including React, TypeScript, Node.js, PostgreSQL, Drizzle ORM, OpenAI GPT-4o, Stripe, WebRTC, and the NMC verification Railway proxy into a unified, coherent, and production-quality application that provides genuine value to its three categories of users.

For patients, PharmaSense eliminates the confusion and uncertainty that surrounds medical prescription interpretation by providing an AI-powered analyser that instantly converts complex medical prescription documents into clear, structured, and actionable health information. The treatment outcome prediction feature goes further by showing patients concretely how their adherence behaviour will affect their recovery, providing a powerful and personalised motivational tool for improving medication compliance. The ability to find and consult NMC-verified doctors through real-time video calls from any location with a suitable internet connection dramatically expands access to qualified medical care, particularly for patients in areas where specialist doctors are not locally available.

For doctors, PharmaSense provides a structured and professional digital consultation environment that handles the administrative complexity of consultation scheduling, payment collection, and medical record management automatically, allowing them to focus their attention on the clinical aspects of their patient interactions. The digital prescription upload feature gives patients immediate access to their

prescription without the delays and limitations of physical documents, and the integrated patient history view provides doctors with the context they need to make informed clinical decisions.

10.2 FUTURE ENHANCEMENTS

Several directions for future development have been identified that would substantially expand the capabilities, reliability, and reach of the PharmaSense platform. The most immediately impactful enhancement would be the development of a dedicated backend NMC verification queue that processes verification requests asynchronously rather than synchronously during doctor registration. Under this architecture, a doctor who registers when the NMC server is slow would receive immediate confirmation that their registration has been received and that NMC verification is in progress, and would then receive a notification when the verification either succeeds or exhausts its attempts and falls back to the manual review queue. This approach would eliminate the current experience where doctors must wait up to several minutes for a synchronous verification attempt that frequently fails due to the NMC server's response time limitations.Push notification support would significantly improve the user experience for both patients and doctors by providing immediate alerts for consultation status changes, appointment reminders, prescription uploads, and medication delivery status updates.

Extending the AI prescription analyser to support prescriptions written in regional Indian languages including Hindi, Tamil, Telugu, Kannada, Marathi, and Bengali would substantially expand the accessibility of the platform for patients whose prescriptions are written in languages other than English.The implementation of end-to-end encryption for all medical records stored in the platform, combined with formal compliance assessment against India's Digital Personal Data Protection Act of 2023, would strengthen the security credentials of the platform and address the growing regulatory expectations around the handling of health-related personal data in India.

APPENDIX – A

SOURCE CODE

nmc.ts

```
const QUICK = { maxRetries: 4, timeoutMs: 30_000, retryDelayMs: 800 };

const THOROUGH = { maxRetries: 6, timeoutMs: 45_000, retryDelayMs: 1_000 };

async function fetchNMCData(

  registrationNumber: string,

  cfg: { maxRetries: number; timeoutMs: number; retryDelayMs: number }

): Promise<NMCFetchResult> {

  for (let attempt = 1; attempt <= cfg.maxRetries; attempt++) {

    try {

      const result = await fetchNMCDataOnce(registrationNumber, cfg.timeoutMs);

      if (result.status !== "unreachable") return result;

      if (attempt < cfg.maxRetries) await sleep(cfg.retryDelayMs);

    } catch {

      logger.info({ attempt }, "NMC fetch timed out, retrying");

      if (attempt < cfg.maxRetries) await sleep(cfg.retryDelayMs);

    }

  }

  logger.info({ registrationNumber }, "NMC server did not respond   proceeding with
pending status");

  return { status: "unreachable" };            }
```

webrtc.ts

```
import { Router } from "express";
import { db, consultationsTable, webrtcSignalsTable } from "@workspace/db";
import { eq, and, asc } from "drizzle-orm";
import { requireAuth } from "../middlewares/auth";
import { randomBytes } from "crypto";

const router = Router();

router.post("/token", requireAuth, async (req, res) => {
  try {
    const { consultationId, role } = req.body;

    if (!consultationId || !role) {
      res.status(400).json({ error: "consultationId and role are required" });
      return;
    }
    const consultations = await db.select().from(consultationsTable)
      .where(eq(consultationsTable.id, consultationId))
      .limit(1);
    if (consultations.length === 0) {
      res.status(404).json({ error: "Consultation not found" });
      return;
```

```
    }
    const consultation = consultations[0];
    const roomId = consultation.roomId || `room_${consultationId}`;
    const token = randomBytes(16).toString("hex");
    res.json({ roomId, token, appId: "mediconnect_webrtc" });
  } catch (err) {
    req.log.error({ err }, "WebRTC token error");
    res.status(500).json({ error: "Internal server error" });
  }
});
router.post("/signal/:consultationId", requireAuth, async (req, res) => {
  try {
    const consultationId = parseInt(req.params.consultationId);
    const { fromRole, toRole, signal } = req.body;
    await db.insert(webrtcSignalsTable).values({
      consultationId,
      fromRole,
      toRole,
      signal: JSON.stringify(signal),
      consumed: false,
    });
    res.json({ success: true });
  } catch (err) {
```

```
    req.log.error({ err }, "WebRTC signal error");
    res.status(500).json({ error: "Internal server error" });
  }
});
router.get("/signal/:consultationId/:role", requireAuth, async (req, res) => {
  try {
    const consultationId = parseInt(req.params.consultationId);
    const role = req.params.role;
    const signals = await db.select()
      .from(webrtcSignalsTable)
      .where(
        and(
          eq(webrtcSignalsTable.consultationId, consultationId),
          eq(webrtcSignalsTable.toRole, role),
          eq(webrtcSignalsTable.consumed, false)
        )
      )
      .orderBy(asc(webrtcSignalsTable.id))
      .limit(20);

    if (signals.length > 0) {
      await db.update(webrtcSignalsTable)
        .set({ consumed: true })
```

```
        .where(
          and(
            eq(webrtcSignalsTable.consultationId, consultationId),
            eq(webrtcSignalsTable.toRole, role)
          )
        );
    }

    res.json(signals.map(s => ({
      id: s.id,
      fromRole: s.fromRole,
      signal: JSON.parse(s.signal),
    })));
  } catch (err) {
    req.log.error({ err }, "WebRTC poll signal error");
    res.status(500).json({ error: "Internal server error" });
  }
});
export default router;
```

APPENDIX – B

SCREENSHOTS

Sample output

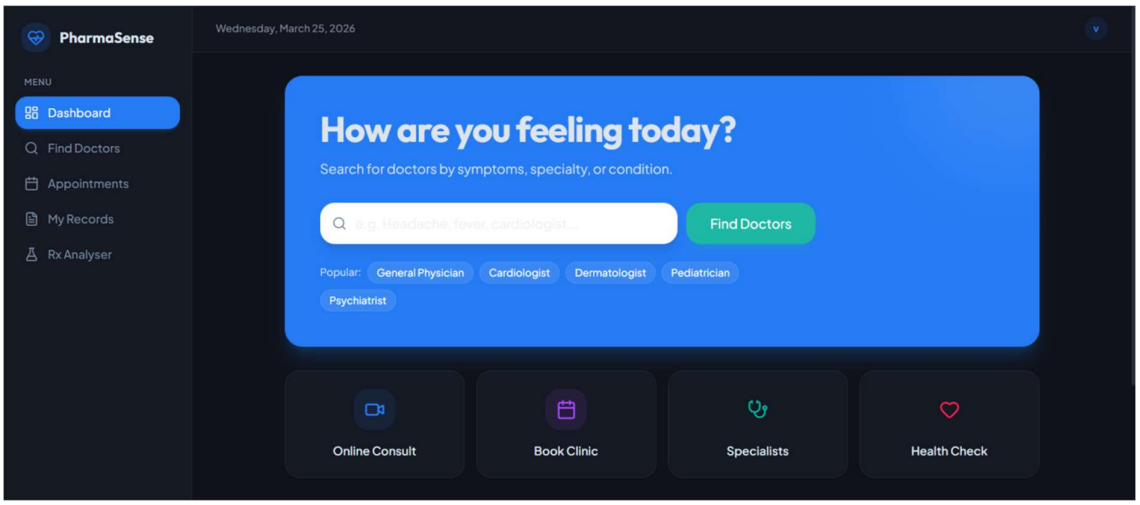

Figure B.1 Patient Dashboard page

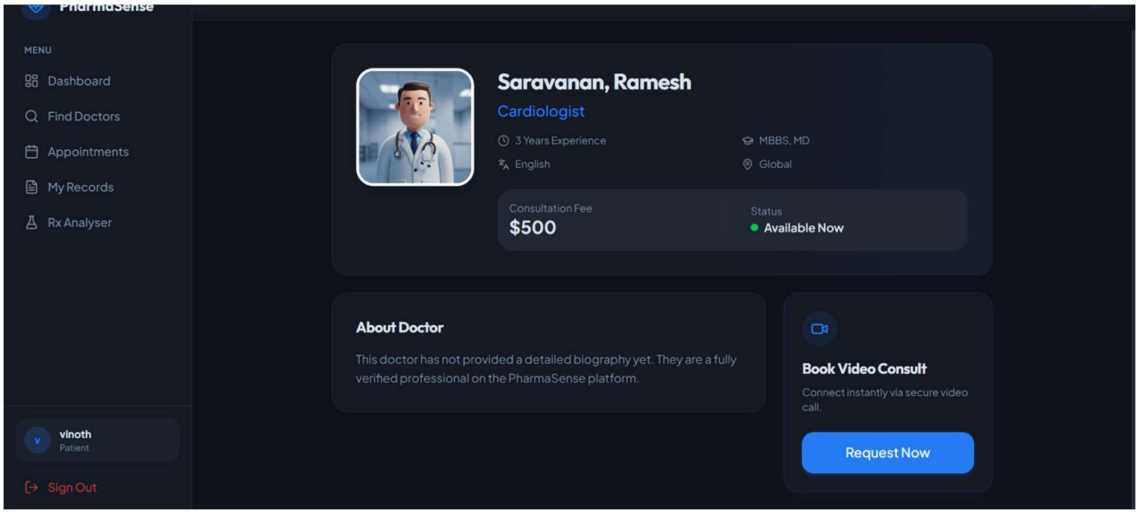

Figure B.2 Doctor Consultation Booking page

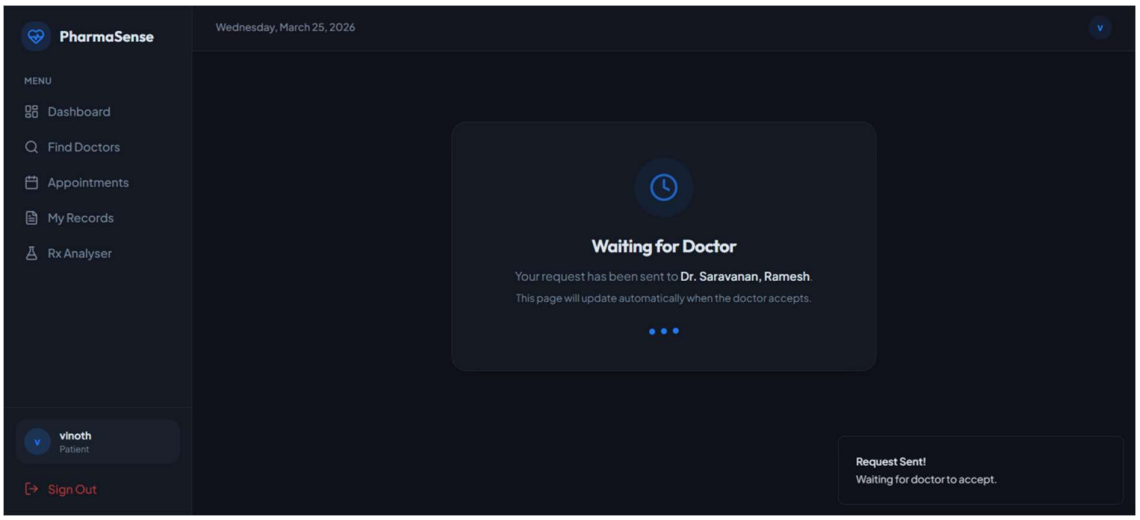

Figure B.3 Doctor Consultation Request page

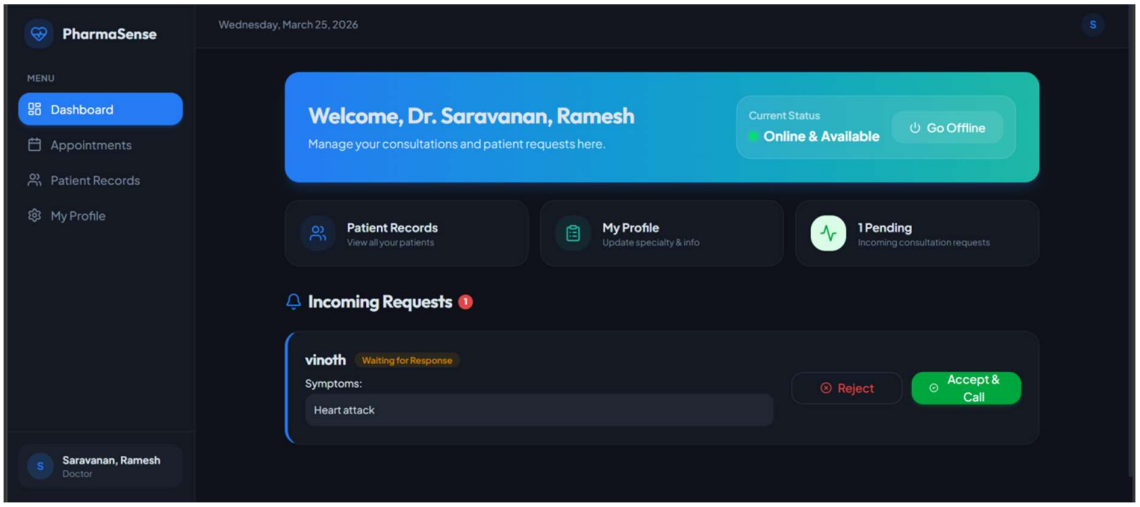

Figure B.4 Doctor Dashboard page

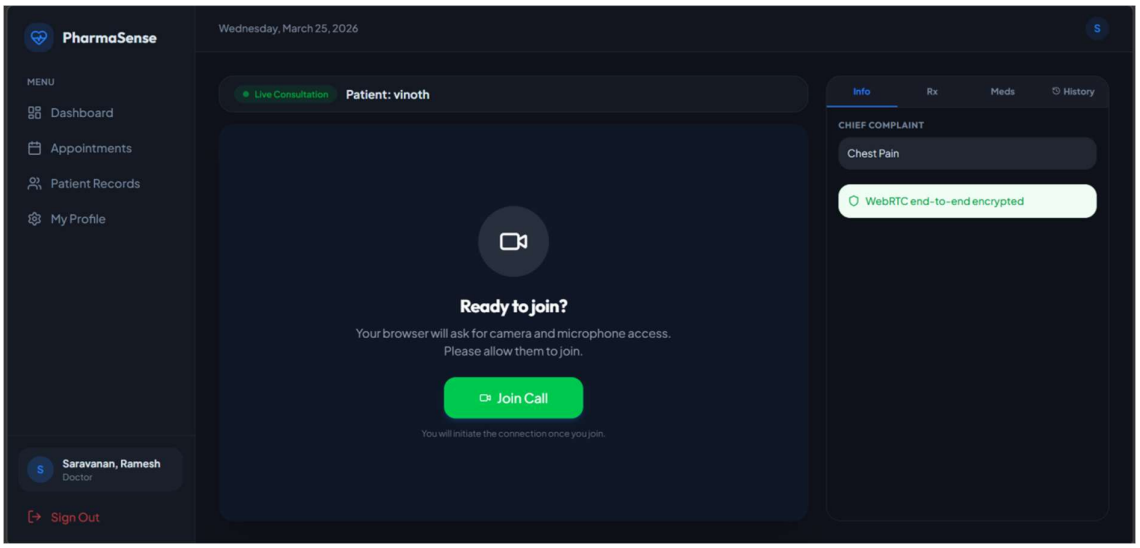

Figure B.5 Doctor Consultation page

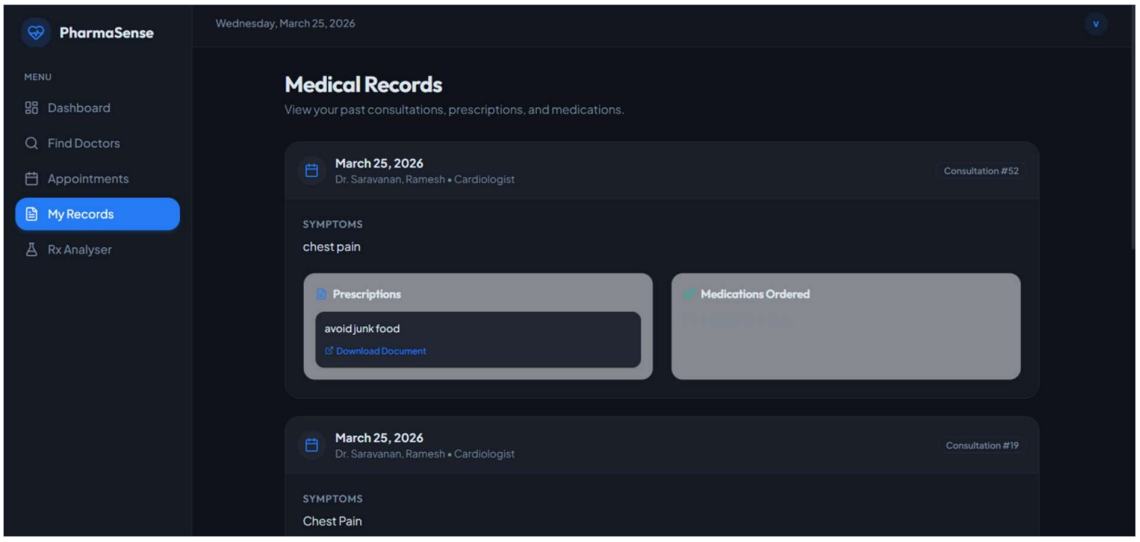

Figure B.6 Patient Medical Records page

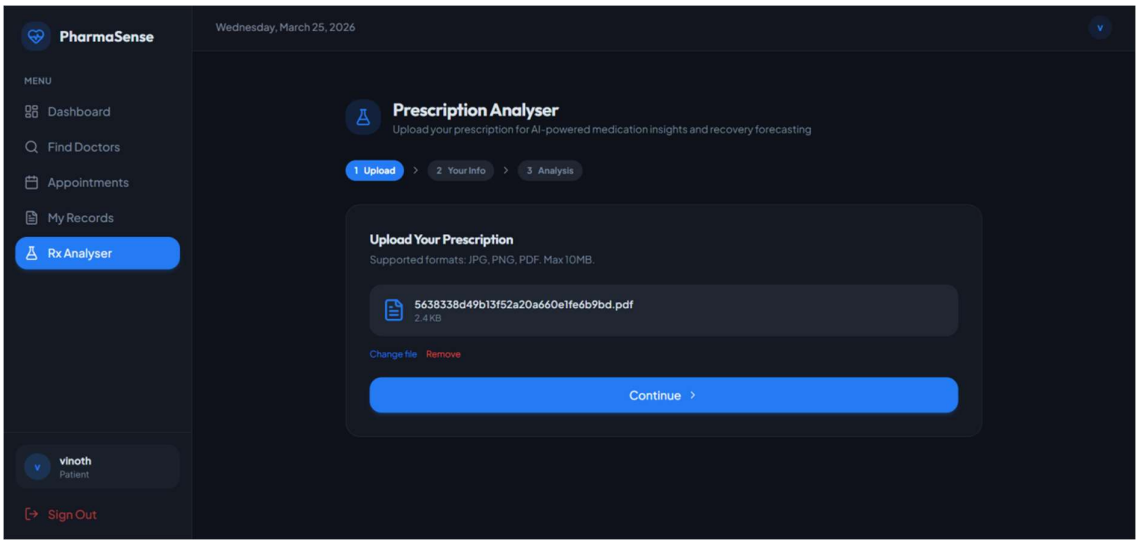

Figure B.7 Prescription Analysis page

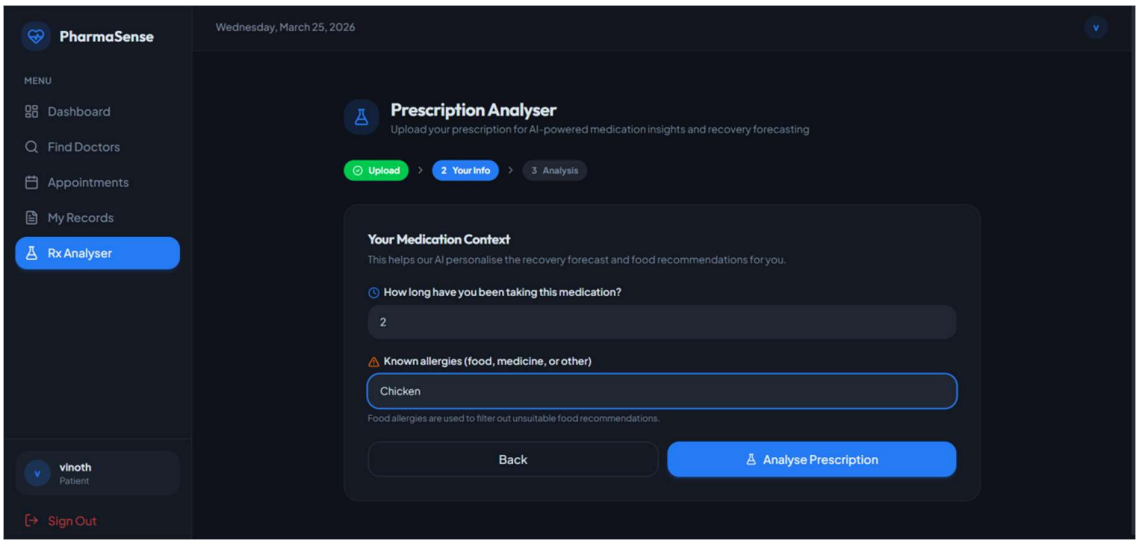

Figure B.8 Medication Context Input page

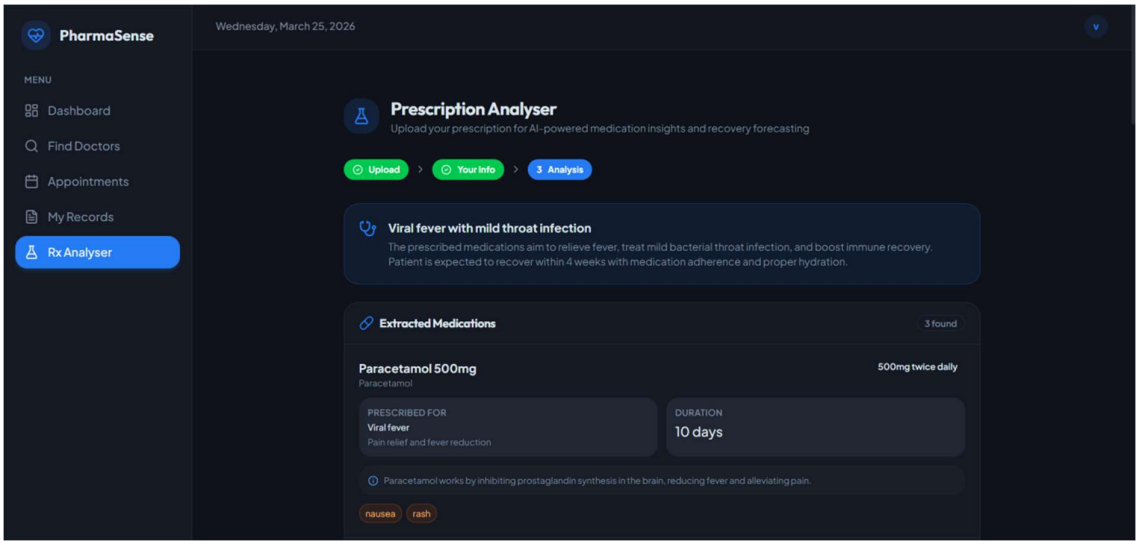

Figure B.9 Extracted Medication Details page

Figure B.10 Recovery Timeline Forecast page

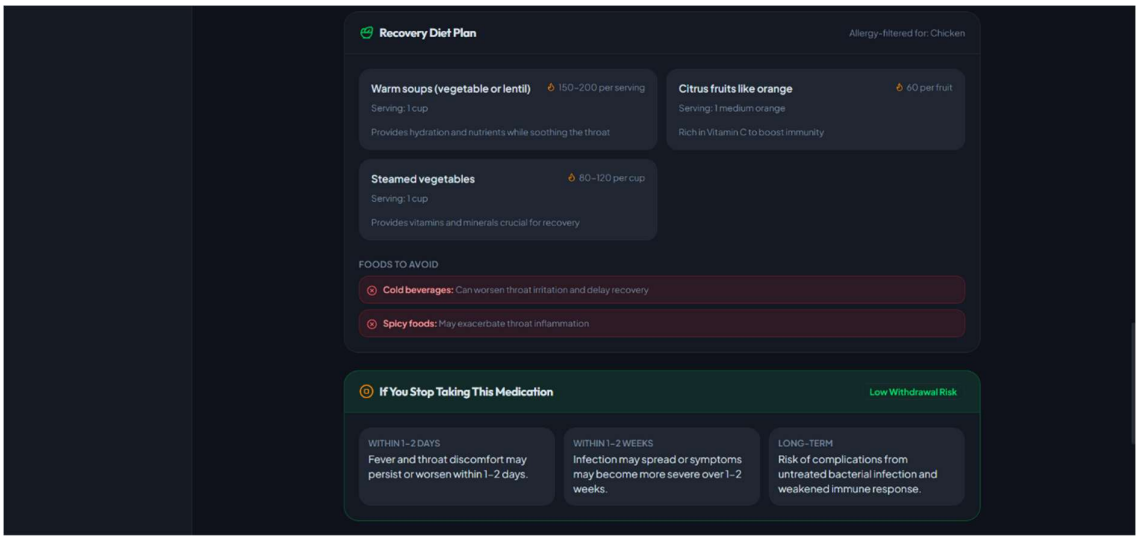

Figure B.11 Recovery Diet Plan page

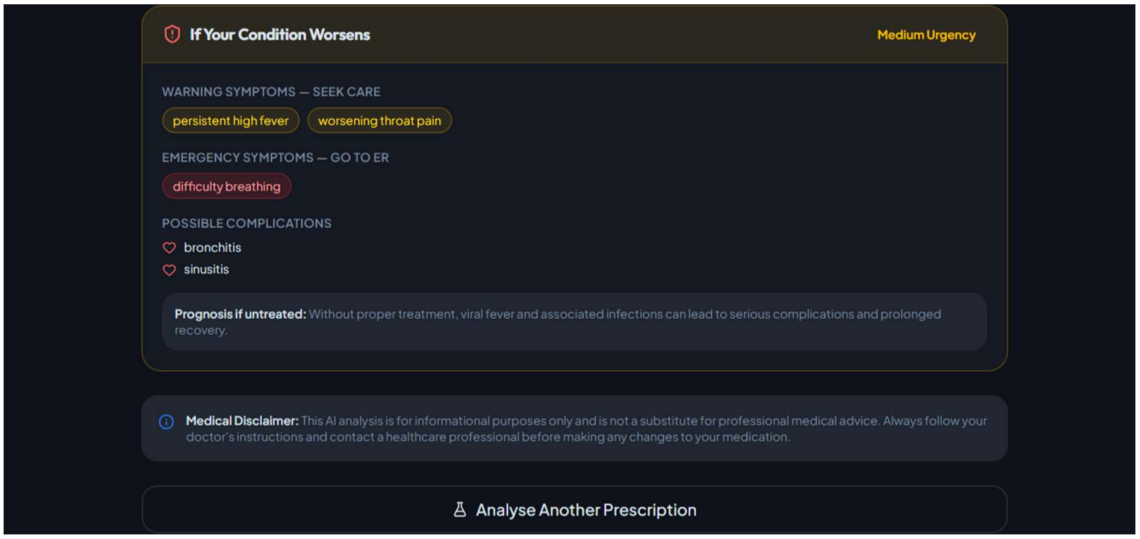

Figure B.12 Worst Case Scenario page

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Certificate of Participation

This is to certify that Mr./Ms. SANTHOSH B from K.RAMAKRISHNAN COLLEGE OF TECHNOLOGY has presented a paper titled PHARMASENSE - AI-BASED PRESCRIPTION ANALYZER AND TREATMENT OUTCOME FORECASTING SYSTEM USING OCR, NLP AND RANDOM FOREST MACHINE LEARNING ALGORITHM in SIXTH INTERNATIONAL CONFERENCE ON ARTIFICIAL INTELLIGENCE AND SMART COMPUTING (ICAISC'26) held during 19th - 21st March 2026 at Bannari Amman Institute of Technology, Sathyamangalam.

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